A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Kinetics

At AS you explained qualitatively why concentration, temperature and catalysts change the rate of a reaction. Here the treatment becomes quantitative: rate equations say exactly how the rate depends on each concentration, the orders they contain reveal which species take part in the slowest step of the mechanism, and the Arrhenius equation puts a number on the activation energy.

Rate of reaction

DEFINITION

The **rate of reaction** is the change in concentration of a reactant or product per unit time. Its usual unit is **mol dm^{-3} s^{-1}**.

On a graph of concentration against time, the rate at any moment is the **gradient** of the curve at that moment. Because the curve bends, you draw a **tangent** to it and find the tangent's gradient. A reactant's concentration falls, so its gradient is negative; the rate is the size of that gradient. The **initial rate** is the gradient of the tangent at t = 0, where the reaction is fastest because the reactants are most concentrated.

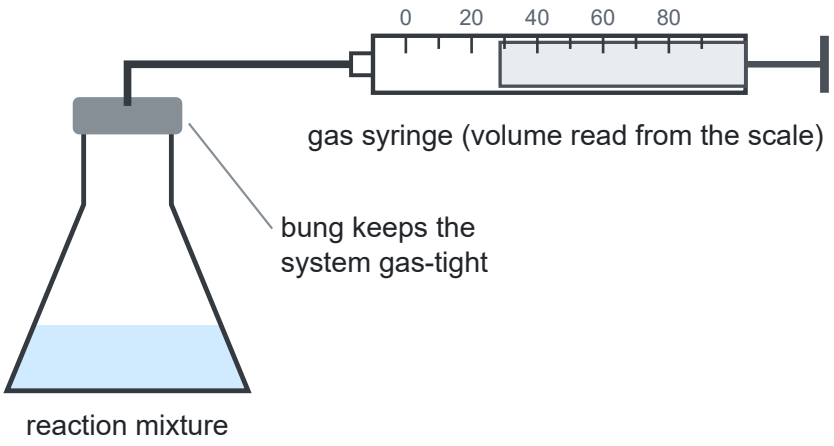
The rate in mol dm^{-3} s^{-1} depends on which substance you follow. In $2N_2O_5 \rightarrow 4NO_2 + O_2$, nitrogen dioxide forms twice as fast as N_2O_5 is used up and four times as fast as oxygen forms, so always say which substance a rate refers to.

Following the rate of a reaction

To follow a reaction you need a property that changes as it goes and that you can measure without disturbing it (or a way of stopping samples so you can analyse them). Choose the method to suit the reaction.

Method	Use it when…	Example
Volume of gas, collected in a gas syringe	a gas is given off	$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$
Loss of mass, on a balance	a gas escapes from an open flask; works best for a dense gas such as CO_2	$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

Colorimetry	a reactant or product is coloured; absorbance is proportional to its concentration	$HCOOH(aq) + Br_2(aq) \rightarrow 2H^+(aq) + 2Br^-(aq) + CO_2(g)$: the orange bromine fades
Titration of quenched samples	an acid, alkali or iodine is used up or formed	$CH_3COCH_3 + I_2 \rightarrow CH_3COCH_2I + H^+ + I^-$: titrate the iodine left with thiosulfate
Electrical conductivity	the number, charge or type of ions changes	$(CH_3)_3CBr + H_2O \rightarrow (CH_3)_3COH + H^+ + Br^-$: ions are produced
"Clock" reaction	you only need the initial rate: time how long a fixed small amount of product takes to form	the iodine clock described in the core practical below



Collecting a gas in a gas syringe. Readings of volume against time give a curve whose gradient is the rate of gas production.

EXAM TIP

Quenching means stopping the reaction in a sample so its composition stays fixed while you analyse it. You can quench by diluting the sample in cold water, or by removing a catalyst: adding sodium hydrogencarbonate neutralises an acid catalyst.

Rate equations and orders

For a reaction between A and B, experiments show that the rate is proportional to each concentration raised to some power:

The rate equation

$$\text{rate} = k[\text{A}]^m[\text{B}]^n$$

$[\text{A}]$: concentration of A / mol dm^{-3} m, n: orders with respect to A and B k: rate constant

- **Order with respect to a reactant:** the power its concentration is raised to in the rate equation. For this course the order is 0, 1 or 2.
- **Overall order:** the sum of the individual orders, m + n.
- **Rate constant, k:** the constant of proportionality. It does not change with concentration, but it is different at each temperature, increasing as the temperature rises. A larger k means a faster reaction.

Order with respect to A	Rate is…	Double [A]	Triple [A]
0	independent of [A] (because $[A]^0 = 1$)	rate × 1	rate × 1
1	proportional to [A]	rate × 2	rate × 3
2	proportional to $[A]^2$	rate × 4	rate × 9

COMMON MISTAKE

Orders can only be found by **experiment**. They are not the numbers in front of the formulae in the balanced equation. For $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$ the rate equation is rate = $k[H_2O_2][I^-]$: first order in iodide despite the 2, and zero order in H^+.

Units of the rate constant

The units of k depend on the overall order, so work them out from the rate equation each time.

Units of k when rate = k[A][B]²

1. Rearrange the rate equation to make k the subject:

$$k = \frac{\text{rate}}{[\text{A}][\text{B}]^2}$$

2 Substitute the units of each quantity:

$$k = \frac{\text{mol}\,\text{dm}^{-3}\,\text{s}^{-1}}{(\text{mol}\,\text{dm}^{-3})(\text{mol}\,\text{dm}^{-3})^2} = \frac{\text{mol}\,\text{dm}^{-3}\,\text{s}^{-1}}{\text{mol}^3\,\text{dm}^{-9}}$$

3 Cancel one mol dm^{-3} from top and bottom, then move what is left on the bottom to the top by changing the signs of its powers:

$$k = \frac{\text{s}^{-1}}{\text{mol}^2\,\text{dm}^{-6}} = \text{dm}^6\,\text{mol}^{-2}\,\text{s}^{-1}$$

The order of the units in the final answer doesn't matter: mol^{-2} dm^6 s^{-1} is equally correct.

Overall order	Example rate equation	Units of k
0	rate = k	mol dm^{-3} s^{-1}
1	rate = k[A]	s^{-1}
2	rate = k[A][B] or $k[A]^2$	dm^3 mol^{-1} s^{-1}
3	rate = $k[A][B]^2$	dm^6 mol^{-2} s^{-1}

Orders from graphs and half-lives

One way to find the order is a **continuous** experiment: follow the concentration of one reactant all the way through a single run. For this to show the order with respect to that reactant alone, every other reactant must be in large excess, so that its concentration hardly changes and has no effect on how the rate changes.

DEFINITION

The **half-life**, $t_{½}$, of a reaction is the time taken for the concentration of a reactant to fall to half its value.

Order	Concentration–time graph	Successive half-lives	Rate–concentration graph
0	straight line sloping down: the rate stays constant until the reactant runs out	each one is half the one before	horizontal line

1	curve that gets less steep (exponential decay)	**constant**, whatever the starting concentration	straight line through the origin
2	steeper at first, then a long, slow tail	each one is double the one before	upward curve through the origin (a straight line if you plot rate against $[A]^2$)

Why is the first-order half-life constant? With rate = k[A], the reaction always uses up the same *fraction* of whatever A is left in a given time. Going from 0.80 to 0.40 mol dm^{-3} takes exactly as long as going from 0.40 to 0.20 mol dm^{-3}: there is half as much to remove, but it is removed at half the rate. Radioactive decay behaves in the same way.

Using a constant half-life

1. A first-order reaction has a half-life of 25 s. The initial concentration of the reactant is 0.64 mol dm^{-3}. What is its concentration after 100 s?
2. The half-life is constant, so count half-lives: 100 s ÷ 25 s = 4 half-lives.
3. Halve the concentration four times:

$$0.64 \rightarrow 0.32 \rightarrow 0.16 \rightarrow 0.080 \rightarrow 0.040 \text{ mol dm}^{-3}$$

Equivalently, $0.64 \times (\frac{1}{2})^4 = 0.64/16 = 0.040 \text{ mol dm}^{-3}$.

For a first-order reaction the half-life and the rate constant are linked by $\ln 2 = k\,t_{1/2}$, so a half-life read from a graph gives k directly. A half-life of 25 s means $k = 0.693/25 = 0.028 \text{ s}^{-1}$. The faster the reaction (the larger k), the shorter the half-life, but it never depends on the starting concentration.

The initial rates method

The other approach is a set of separate experiments, each started with different initial concentrations, measuring only the initial rate each time (from the gradient of a tangent at t = 0, or from a clock reaction). Change **one** concentration at a time and keep the others the same, so any change in rate is caused by that reactant alone.

Experiment	[NO] / mol dm^{-3}	$[H_2]$ / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	5.0×10^{-3}	2.0×10^{-3}	1.25×10^{-5}
2	1.0×10^{-2}	2.0×10^{-3}	5.0×10^{-5}

3	1.0×10^{-2}	4.0×10^{-3}	1.0×10^{-4}

Rate equation for $2NO(g) + 2H_2(g) \rightarrow N_2(g) + 2H_2O(g)$

1. **Order with respect to NO:** compare experiments 1 and 2, where only [NO] changes. [NO] doubles and the rate goes from 1.25×10^{-5} to 5.0×10^{-5}, a factor of $4 = 2^2$.

   So the reaction is second order with respect to NO.

2. **Order with respect to H_2:** compare experiments 2 and 3, where only $[H_2]$ changes. $[H_2]$ doubles and the rate doubles.

   So the reaction is first order with respect to H_2.

3. Write the rate equation. The overall order is 2 + 1 = 3.

$$\text{rate} = k[\text{NO}]^2[\text{H}_2]$$

4. Find k by substituting the values from any one experiment, here experiment 1:

$$k = \frac{\text{rate}}{[\text{NO}]^2[\text{H}_2]} = \frac{1.25 \times 10^{-5}}{(5.0 \times 10^{-3})^2 \times 2.0 \times 10^{-3}} = 250\ \text{dm}^6\,\text{mol}^{-2}\,\text{s}^{-1}$$

   Check with another experiment: $250 \times (1.0 \times 10^{-2})^2 \times 4.0 \times 10^{-3} = 1.0 \times 10^{-4}$, *matching experiment 3.*

When two concentrations change at once

Sometimes no pair of experiments changes only one concentration. The effects then **multiply**. Use an order you already know to strip out its effect, and whatever factor is left belongs to the other reactant.

Experiment	[P] / mol dm^{-3}	[Q] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.020	0.030	4.0×10^{-4}
2	0.040	0.030	8.0×10^{-4}
3	0.060	0.090	1.08×10^{-2}

Untangling two changes at once

1. Experiments 1 and 2: only [P] changes. It doubles and the rate doubles, so the reaction is **first order in P**.

2. Experiments 1 and 3: both change. The rate rises by a factor of $1.08 \times 10^{-2} / 4.0 \times 10^{-4} = 27$.

3. [P] has tripled (0.020 → 0.060). Being first order, that alone triples the rate. The rest of the increase must come from Q: $27 \div 3 = 9$.

4. [Q] has also tripled (0.030 → 0.090) and multiplied the rate by 9 = 3^2, so the reaction is **second order in Q**.

$$\text{rate} = k[\text{P}][\text{Q}]^2, \qquad k = \frac{4.0 \times 10^{-4}}{0.020 \times (0.030)^2} = 22 \text{ dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$$

Rate equations and mechanisms

Most reactions do not happen in a single collision. They go through a sequence of simple steps called the **mechanism**, often forming **intermediates**: species made in one step and used up in a later one. Adding the steps together, and cancelling the intermediates, must give the overall equation.

DEFINITION

The **rate-determining step** is the slowest step in the mechanism. The overall reaction cannot go faster than this step, just as a queue cannot move faster than its slowest checkout.

- A species that appears in the rate equation takes part in the rate-determining step, or in a step before it that makes an intermediate used in the rate-determining step.
- Its order tells you how many particles of it are involved in reaching and completing the rate-determining step.
- A reactant that is **zero order** reacts only **after** the rate-determining step. So if any reactant is zero order, the mechanism must have at least two steps.
- An intermediate never appears in the rate equation. Replace it with the reactants that formed it.
- A catalyst can appear in the rate equation, even though it is not in the overall equation, if it takes part in the rate-determining step.

Which step is rate-determining?

1. For $2NO(g) + Br_2(g) \rightarrow 2NOBr(g)$ the rate equation is rate = $k[NO]^2[Br_2]$. A two-step mechanism is proposed:

   Step 1: $NO + Br_2 \rightarrow NOBr_2$
   Step 2: $NOBr_2 + NO \rightarrow 2NOBr$

2. Check the steps add up: $NO + Br_2 + NOBr_2 + NO \rightarrow NOBr_2 + 2NOBr$. Cancelling $NOBr_2$ leaves $2NO + Br_2 \rightarrow 2NOBr$, so $NOBr_2$ is an intermediate.

3. If step 1 were the slow step, it would contain one NO and one Br_2, giving rate = $k[NO][Br_2]$. That doesn't match the experiment.

4. If step 2 is slow, it uses one $NOBr_2$ (made from one NO and one Br_2) and one more NO: two NO and one Br_2 in total. That gives rate = $k[NO]^2[Br_2]$, matching the experiment.

So step 2 is the rate-determining step, and step 1 must be fast.

Overall reaction and rate equation	Mechanism consistent with it
$NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$ rate = $k[NO_2]^2$	$2NO_2 \rightarrow NO_3 + NO$ (slow) $NO_3 + CO \rightarrow NO_2 + CO_2$ (fast) CO is zero order because it only reacts after the slow step.
$H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$ rate = $k[H_2O_2][I^-]$	$H_2O_2 + I^- \rightarrow IO^- + H_2O$ (slow) $IO^- + H^+ \rightarrow HIO$ (fast) $HIO + H^+ + I^- \rightarrow I_2 + H_2O$ (fast)
$(CH_3)_3CBr + OH^- \rightarrow (CH_3)_3COH + Br^-$ rate = $k[(CH_3)_3CBr]$	$(CH_3)_3CBr \rightarrow (CH_3)_3C^+ + Br^-$ (slow) $(CH_3)_3C^+ + OH^- \rightarrow (CH_3)_3COH$ (fast): S_N1
$CH_3CH_2Br + OH^- \rightarrow CH_3CH_2OH + Br^-$ rate = $k[CH_3CH_2Br][OH^-]$	a single step in which OH^- attacks as Br^- leaves: S_N2

KEY IDEA

Kinetics is how chemists tell S_N1 from S_N2. If the rate of hydrolysis depends on $[OH^-]$, the hydroxide ion must be in the rate-determining step, so the mechanism is the one-step S_N2 route (primary

halogenoalkanes). If changing [OH^-] has no effect, the slow step is the halogenoalkane ionising on its own: S_N1 (tertiary halogenoalkanes). See the halogenoalkanes page for the mechanisms.

Core practicals: following rates in the lab

CORE PRACTICAL

The iodine–propanone reaction, followed by titration

Propanone reacts with iodine in acid solution; the acid is a catalyst:

$$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$$

Mix propanone and sulfuric acid, both in large excess, then add iodine solution and start the clock. At regular intervals withdraw a 10.0 cm^3 sample with a pipette and run it into excess sodium hydrogencarbonate solution. This **quenches** the sample by neutralising the acid catalyst. Titrate the iodine left in the sample with sodium thiosulfate solution, adding starch near the end point (blue-black to colourless):

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

- The titre is proportional to [I_2]. A graph of titre against time is a **straight line**, so the rate is constant while iodine is used up: the reaction is **zero order with respect to iodine**.
- Repeating with different concentrations of propanone or acid changes the gradient. These experiments show it is first order in propanone and first order in H^+: rate = $k[CH_3COCH_3][H^+]$.
- So the rate-determining step involves one propanone molecule and one H^+ ion, and iodine reacts in a later, fast step. H^+ appears in the rate equation even though it is a catalyst.
- The same reaction can be followed by colorimetry, since iodine is the only coloured species.
- **Safety:** propanone is highly flammable; iodine solution stains and irritates; wear eye protection.

CORE PRACTICAL

A clock reaction: initial rates and activation energy

Hydrogen peroxide oxidises iodide ions in acid: $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$. A small, fixed amount of sodium thiosulfate and some starch are added. Each iodine molecule formed is turned straight back into iodide by the thiosulfate, until the thiosulfate is used up. Then iodine builds up and the starch suddenly turns blue-black. Record the time, t.

- Every run makes the same small amount of iodine before the colour appears, so the average rate up to that point is proportional to **1/t**.
- The thiosulfate is kept small so that only a tiny fraction of the reactants is used before the colour appears. Their concentrations have barely changed, so 1/t is a good measure of the **initial** rate.
- To find an order, change the concentration of one reactant at a time (making up the volume with water so the total volume is constant) and plot 1/t against concentration.

- To find the activation energy, keep every concentration the same and repeat at several temperatures in a water bath, letting the separate solutions reach the bath temperature before mixing. Since k ∝ 1/t, a graph of ln(1/t) against 1/T has gradient $-E_a/R$ (see below).

Temperature and the Arrhenius equation

Raising the temperature increases the rate constant. The main reason is the shape of the Maxwell–Boltzmann distribution: at a higher temperature a much larger fraction of collisions have at least the activation energy, E_a. (The molecules also collide a little more often, but that is a far smaller effect.) The Arrhenius equation describes this exactly:

The Arrhenius equation and its logarithmic form

$$k = A\,e^{-E_a/RT} \qquad \ln k = \ln A - \frac{E_a}{R}\cdot\frac{1}{T}$$

k: rate constant A: Arrhenius constant (same units as k) E_a: activation energy / J mol^{-1}
R = 8.31 J K^{-1} mol^{-1} T: temperature / K

The factor $e^{-E_a/RT}$ is the fraction of collisions with enough energy to react. It is tiny when E_a is large or T is small, and it grows very rapidly with temperature. For a typical activation energy of about 50 kJ mol^{-1}, warming from 298 K to 308 K roughly doubles the rate constant.

COMMON MISTAKE

E_a is usually quoted in kJ mol^{-1}, but R is in J K^{-1} mol^{-1}. Multiply E_a by 1000 before substituting, and use T in kelvin.

Finding E_a from a graph

Compare the logarithmic form with the equation of a straight line, y = mx + c. Plotting **ln k** (y) against **1/T** (x) gives a straight line with **gradient $-E_a/R$** and intercept ln A. The gradient is always negative, and the steeper it is, the larger the activation energy. Because the rate is proportional to k at fixed concentrations, ln(rate) or ln(1/t) from a clock reaction can be plotted instead of ln k: only the intercept changes, not the gradient.

T / K	t / s	1/T / 10^{-3} K^{-1}	ln(1/t)
293	240	3.413	−5.48
303	115	3.300	−4.74

313	58.2	3.195	−4.06
323	30.6	3.096	−3.42
333	16.7	3.003	−2.82

Activation energy from clock-reaction times

1. Calculate 1/T and ln(1/t) for each run (table above) and plot ln(1/t) against 1/T. The points lie on a straight line. Draw the line of best fit.

2. Find the gradient using two points on the line that are far apart, here the first and last:

$$\text{gradient} = \frac{-2.82 - (-5.48)}{(3.003 - 3.413) \times 10^{-3}} = \frac{2.66}{-0.410 \times 10^{-3}} = -6490 \text{ K}$$

3. The gradient equals $-E_a/R$, so:

$$E_a = -\text{gradient} \times R = 6490 \times 8.31 = 53\,900 \text{ J mol}^{-1} = 53.9 \text{ kJ mol}^{-1}$$

The gradient has units of K because ln(1/t) has no units and 1/T is in K^{-1}. Multiplying K by J K^{-1} mol^{-1} gives J mol^{-1}.

Calculating the Arrhenius constant A

1. A first-order reaction has k = 2.5 × 10^{-4} s^{-1} at 320 K and E_a = 85.0 kJ mol^{-1}. Calculate A.

2. Work out the exponent, with E_a in J mol^{-1}:

$$\frac{E_a}{RT} = \frac{85\,000}{8.31 \times 320} = 31.96 \qquad e^{-31.96} = 1.31 \times 10^{-14}$$

3. Rearrange $k = A\,e^{-E_a/RT}$:

$$A = \frac{k}{e^{-E_a/RT}} = \frac{2.5 \times 10^{-4}}{1.31 \times 10^{-14}} = 1.9 \times 10^{10} \text{ s}^{-1}$$

A has the same units as k, here s^{-1}.

How much faster does a catalyst make a reaction?

1. A catalyst lowers the activation energy of a reaction from 75 kJ mol^{-1} to 50 kJ mol^{-1}. Assuming A is unchanged, by what factor does k increase at 298 K?

2. Divide the two Arrhenius expressions. A cancels:

$$\frac{k_{\text{cat}}}{k} = \frac{e^{-50\,000/RT}}{e^{-75\,000/RT}} = e^{25\,000/(8.31\times298)} = e^{10.1} \approx 2.4 \times 10^{4}$$

Lowering E_a by a third makes the reaction about 24 000 times faster, which is why catalysts matter so much in industry.

Catalysts

A catalyst speeds up a reaction without being used up, by providing an alternative route (mechanism) with a lower activation energy. It speeds up the forward and reverse reactions equally, so it does not change the position of equilibrium or the enthalpy change.

- **Homogeneous catalyst:** in the same phase as the reactants, usually all in solution. It reacts to form an intermediate and is regenerated in a later step. Because it takes part in the mechanism it can appear in the rate equation: H^+ in the iodination of propanone is an example.
- **Heterogeneous catalyst:** in a different phase from the reactants, usually a solid with gaseous or dissolved reactants. The reaction happens on its surface.

How a heterogeneous catalyst works

Reactant molecules **adsorb** (form weak bonds) onto **active sites** on the catalyst surface. This weakens bonds within the reactants, holds them close together in a favourable orientation, and concentrates them on the surface, so they react more easily. The products then **desorb**, freeing the sites for more reactant. A large surface area, from a fine powder or a thin coating on a support, gives more active sites.

The number of active sites is limited. At high pressure or concentration every site can be occupied, and adding more reactant then cannot speed the reaction up. The rate is then independent of that reactant's concentration: the reaction is zero order with respect to it. The decomposition of ammonia on a hot tungsten surface behaves like this.

Quick checks

QUICK CHECK

The rate equation for a reaction is rate = $k[X]^2[Y]$. What are the units of k?

- $mol^2\ dm^{-6}\ s^{-1}$
- $dm^3\ mol^{-1}\ s^{-1}$
- $dm^6\ mol^{-2}\ s^{-1}$
- $mol\ dm^{-3}\ s^{-1}$

The overall order is 3. k = rate/($[X]^2[Y]$) = (mol dm^{-3} s^{-1})/(mol $dm^{-3})^3$ = mol^{-2} dm^6 s^{-1}. The second option is for a second-order reaction.

QUICK CHECK

A reaction is first order with respect to A and second order with respect to B. Both concentrations are doubled at the same time. By what factor does the rate increase?

- 3
- 4
- 6
- 8

Doubling [A] multiplies the rate by 2; doubling [B] multiplies it by 2^2 = 4. The effects multiply: 2 × 4 = 8. Adding the orders (to get 3) or adding the factors (to get 6) are the usual slips.

QUICK CHECK

A first-order reaction has a half-life of 1400 s. Starting from 1.0 mol dm^{-3}, what is the concentration after 4200 s?

- 0.33 mol dm^{-3}
- 0.125 mol dm^{-3}
- 0.25 mol dm^{-3}
- 0.50 mol dm^{-3}

4200 s is three half-lives, because a first-order half-life is constant: 1.0 → 0.50 → 0.25 → 0.125 mol dm^{-3}. Dividing by 3 instead of halving three times gives the wrong answer of 0.33.

QUICK CHECK

The iodination of propanone has rate = k[CH_3COCH_3][H^+]. Which statement must be true?

- Iodine is a catalyst for the reaction.
- Iodine reacts in a step after the rate-determining step.
- The reaction has a single-step mechanism.
- Doubling [I_2] halves the rate.

Iodine is a reactant, but it is zero order, so it is not involved in the slow step and must react afterwards. A zero-order reactant means there are at least two steps. Changing [I_2] has no effect on the rate.

QUICK CHECK

A graph of ln k against 1/T for a reaction has a gradient of −7500 K. What is the activation energy? (R = 8.31 J K^{-1} mol^{-1})

- [x] 62.3 kJ mol^{-1}
- [ ] −62.3 kJ mol^{-1}
- [ ] 902 kJ mol^{-1}
- [ ] 0.90 kJ mol^{-1}

Gradient = $-E_a/R$, so E_a = 7500 × 8.31 = 62 300 J mol^{-1} = 62.3 kJ mol^{-1}. Activation energy is always positive; dividing by R instead of multiplying gives 0.90.

QUICK CHECK

In a clock reaction, why can 1/t be used as a measure of the initial rate?

- [ ] Because the reaction is zero order, so its rate never changes.
- [ ] Because the thiosulfate is a catalyst that keeps the rate constant.
- [x] Because the same small amount of product forms in every run, before the reactant concentrations have changed much.
- [ ] Because the colour appears when half of the reactants have been used up.

Rate = amount formed ÷ time. The amount (fixed by the thiosulfate) is the same in every run, so the rate is proportional to 1/t. Keeping that amount small means the concentrations are still close to their starting values, so it is an initial rate.

QUICK CHECK

Increasing the pressure of a gas reacting on a solid catalyst has no effect on the rate. What is the best explanation?

- The catalyst has been poisoned.
- The activation energy has risen to compensate.
- Gases do not adsorb onto solid surfaces.
- All the active sites on the catalyst surface are already occupied.

Reaction only happens at active sites. Once they are saturated, extra gas molecules have nowhere to adsorb, so the rate stays the same and the reaction is zero order in that gas.

Entropy and energetics

Why does sodium chloride exist, why is it $CaCl_2$ and not CaCl, and why do some reactions that take in heat still happen by themselves? This topic answers with two tools: Born–Haber and solution cycles, which measure the energy holding ionic solids together, and entropy, which counts the ways energy can be shared out and decides whether a change is feasible at all.

The enthalpy changes behind an ionic lattice

The **lattice energy** of an ionic compound measures how strongly its ions hold each other. It cannot be measured directly, because you cannot collect a mole of free gaseous ions and let them crash together. Instead it is calculated with Hess's law from enthalpy changes that *can* be measured. You need to know each of them precisely. All are standard values: measured at 100 kPa and a stated temperature, usually 298 K, with substances in their standard states.

Enthalpy change	Definition	Example	Sign
Enthalpy change of formation, $\Delta_f H$	1 mol of a compound forms from its elements in their standard states	$Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$	usually −
Enthalpy change of atomisation, $\Delta_{at} H$	1 mol of **gaseous atoms** forms from the element in its standard state	$Na(s) \rightarrow Na(g)$ $\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$	always +
First ionisation energy	1 mol of electrons is removed from 1 mol of gaseous atoms, giving 1 mol of gaseous 1+ ions	$Mg(g) \rightarrow Mg^+(g) + e^-$	always +
Second ionisation energy	1 mol of electrons is removed from 1 mol of gaseous 1+ ions, giving 1 mol of gaseous 2+ ions	$Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$	always +
First electron affinity	1 mol of gaseous atoms each gain one electron, giving 1 mol of gaseous 1− ions	$Cl(g) + e^- \rightarrow Cl^-(g)$	− for these elements
Second electron affinity	1 mol of gaseous 1− ions each gain one electron, giving 1 mol of gaseous 2− ions	$O^-(g) + e^- \rightarrow O^{2-}(g)$	+
Lattice energy, LE	1 mol of an ionic solid forms from its **gaseous ions**	$Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$	always −

- **Why is the first electron affinity negative?** The incoming electron is attracted by the nucleus of the neutral atom, so energy is released. Going down group 7 from chlorine to iodine it becomes less exothermic: the atoms are bigger, with more shielding, so the added electron is held less strongly.
- **Why is the second electron affinity positive?** The electron is being added to an ion that is already negative. Energy has to be put in to overcome the repulsion between them. For oxygen it is +798 kJ mol^{-1}, much larger than the −141 kJ mol^{-1} released by the first.
- **Atomisation of a diatomic element** is per mole of atoms formed, so $\Delta_{at}H(Cl)$ is half the Cl–Cl bond enthalpy: about +122 kJ mol^{-1} from a bond enthalpy of about +243 kJ mol^{-1}.

COMMON MISTAKE

Lattice energy is defined here as the enthalpy change for **forming** the solid from gaseous ions, so it is negative. The reverse process, breaking the lattice apart, has the same size but a positive sign. Read every question carefully to see which direction it means, and always write state symbols: the ions must be gaseous.

Born–Haber cycles

A Born–Haber cycle is a Hess's law cycle that links the formation of an ionic solid directly from its elements with an indirect route through gaseous atoms and gaseous ions. Both routes start and finish in the same place, so their total enthalpy changes are equal:

Born–Haber cycle for a compound MX

$$\Delta_f H = \Delta_{at} H(\mathrm{M}) + \mathrm{IE}(\mathrm{M}) + \Delta_{at} H(\mathrm{X}) + \mathrm{EA}(\mathrm{X}) + \mathrm{LE}(\mathrm{MX})$$

IE: ionisation energy (or energies) EA: electron affinity (or affinities) LE: lattice energy

Lattice energy of sodium chloride

1. Data / kJ mol^{-1}: $\Delta_f H(NaCl) = -411$; $\Delta_{at}H(Na) = +107$; first IE(Na) = +496; $\Delta_{at}H(Cl) = +122$; first EA(Cl) = −349.

2. Go round the indirect route from the elements to the gaseous ions: atomise sodium, ionise it, atomise chlorine, add the electron to chlorine.

$$107 + 496 + 122 + (-349) = +376 \ \mathrm{kJ\,mol^{-1}}$$

So $Na^+(g)$ + $Cl^-(g)$ sits 376 kJ mol^{-1} above the elements.

3 Hess's law: $\Delta_f H = 376 + \text{LE}$, so

$$\text{LE} = \Delta_f H - 376 = -411 - 376 = -787 \text{ kJ mol}^{-1}$$

- **Group 2 halides such as $MgCl_2$:** include both the first and second ionisation energies of the metal, and **twice** the atomisation enthalpy and electron affinity of chlorine, because there are two chloride ions. ($2 \times \Delta_{at}H(Cl)$ equals the Cl–Cl bond enthalpy.)
- **Oxides such as MgO:** include the first *and second* electron affinities of oxygen. The second is endothermic, so its arrow points **up** in the cycle.
- Any one value in the cycle can be found if all the others are known: exam questions often ask for an electron affinity or an atomisation enthalpy instead of the lattice energy.

EXAM TIP

When drawing a cycle, put the gaseous ions at the top, with every species and state symbol on each level, and include the electrons (e^-) on the levels where they are free. Up arrows are endothermic and down arrows exothermic.

Why calcium chloride is $CaCl_2$

Born–Haber cycles can test compounds that don't exist. A theoretical lattice energy can be estimated for CaCl and $CaCl_3$ from the sizes and charges of the ions, and the cycle then predicts their enthalpies of formation.

Compound	Ionisation energies needed / kJ mol^{-1}	Lattice energy / kJ mol^{-1}	$\Delta_f H$ / kJ mol^{-1}
CaCl	590	about −720 (estimated)	about −180
$CaCl_2$	590 + 1145 = 1735	−2255 (from the Born–Haber cycle)	−796 (measured)
$CaCl_3$	590 + 1145 + 4912 = 6647	about −4600 (estimated)	about +1540

The enthalpy of formation is largely a contest between the energy needed to make the cation and the energy released when the lattice forms. Going from Ca^+ to Ca^{2+} costs an extra 1145 kJ mol^{-1}, but the doubly charged ion attracts two chloride ions far more strongly, and the lattice energy grows by over 1500 kJ mol^{-1}: $CaCl_2$ is much more exothermic than CaCl. Removing a third electron would mean breaking into the full inner shell of calcium, which costs 4912 kJ mol^{-1}. The extra lattice energy cannot repay that, so $CaCl_3$ would be strongly endothermic. Even CaCl, though exothermic, would not survive: $2CaCl(s) \rightarrow Ca(s) + CaCl_2(s)$ releases about 440 kJ mol^{-1}.

What makes a lattice energy large?

Lattice energy comes from the electrostatic attraction between oppositely charged ions. That attraction is stronger when the charges are larger and when the ions can get closer together. So lattice energy becomes more exothermic when:

- **the ions are smaller:** their centres are closer, so they attract more strongly. Lattice energies become less negative down a group.
- **the ions have larger charges:** the attraction depends on the product of the charges, so going from 1+ and 1− ions to 2+ and 2− ions roughly quadruples it.

Compound	LE / kJ mol^{-1}	What it shows
NaF, NaCl, NaBr, NaI	−928, −787, −751, −703	bigger anion, less exothermic
NaCl, KCl	−787, −718	bigger cation, less exothermic
$MgCl_2$, $CaCl_2$	−2524, −2255	2+ ions give much larger values; Mg^{2+} is smaller than Ca^{2+}
NaF, MgO	−928, −3845	similar-sized ions, but charges 1+/1− against 2+/2−: about four times larger

Theoretical and experimental lattice energies

A lattice energy can also be calculated purely from theory, using the **perfect ionic model**: the ions are treated as perfect spheres with their charge spread evenly, held together only by electrostatic attraction. Comparing this theoretical value with the experimental value from a Born–Haber cycle tests whether the model is right.



purely ionic: both ions spherical

polarised: electron density pulled towards the cation (covalent character

A small, highly charged cation distorts the electron cloud of a large anion, so some electron density is shared between the ions.

- If the two values agree closely, the compound is almost **purely ionic**.

- If the Born–Haber value is **more exothermic** than the theoretical one, the bonding has some **covalent character**. The cation pulls the anion's electron cloud towards itself (**polarisation**), concentrating electron density between the ions; this extra attraction makes the real lattice stronger than the model predicts.
- **Polarising power** of a cation is greatest when it is small and highly charged (high charge density): Mg^{2+} polarises more than Na^{+}, and Al^{3+} more still.
- **Polarisability** of an anion is greatest when it is large (and highly charged): its outer electrons are far from the nucleus and easily pulled. I^{-} is far more polarisable than F^{-}.

Compound	Born–Haber LE / kJ mol^{-1}	Theoretical LE / kJ mol^{-1}	Difference
NaCl	about −787	about −770	about 2%: nearly purely ionic
AgI	about −890	about −780	about 12%: large, polarisable iodide
MgI_2	about −2330	about −1940	about 17%: small 2+ cation and large anion

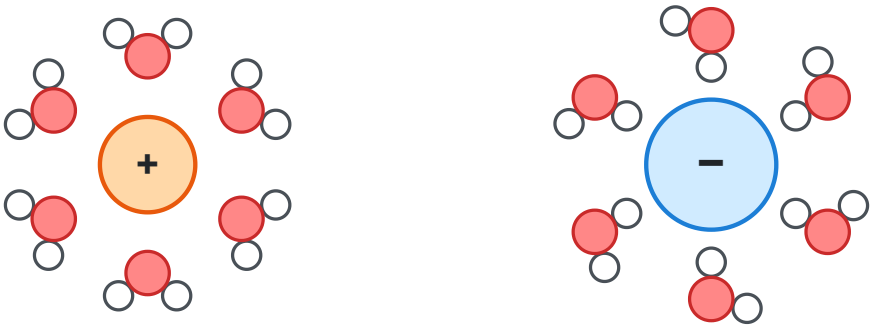
Enthalpies of solution and hydration

DEFINITION

The **enthalpy change of solution**, $\Delta_{sol}H$, is the enthalpy change when 1 mol of an ionic solid dissolves in enough water that the ions are far apart and do not interact: $NaCl(s) + aq \rightarrow Na^{+}(aq) + Cl^{-}(aq)$.

The **enthalpy change of hydration**, $\Delta_{hyd}H$, is the enthalpy change when 1 mol of gaseous ions is dissolved in enough water that further dilution causes no further enthalpy change: $Na^{+}(g) + aq \rightarrow Na^{+}(aq)$.

Water molecules are polar. The δ− oxygen atoms cluster round positive ions and the δ+ hydrogen atoms point at negative ions. These ion–dipole attractions release energy, so **hydration enthalpies are always negative**.



δ− oxygen atoms face the cation

δ+ hydrogen atoms face the anion

Hydrated ions. Oxygen atoms are red and hydrogen atoms white.

Hydration enthalpy is more exothermic for ions with a higher **charge density**: a smaller radius or a larger charge lets the water molecules get closer and attract more strongly.

Ion	Li^+	Na^+	K^+	Mg^{2+}	Ca^{2+}	F^-	Cl^-	Br^-	I^-
$\Delta_{hyd}H$ / kJ mol^{-1}	−519	−406	−322	−1920	−1650	−506	−364	−335	−293

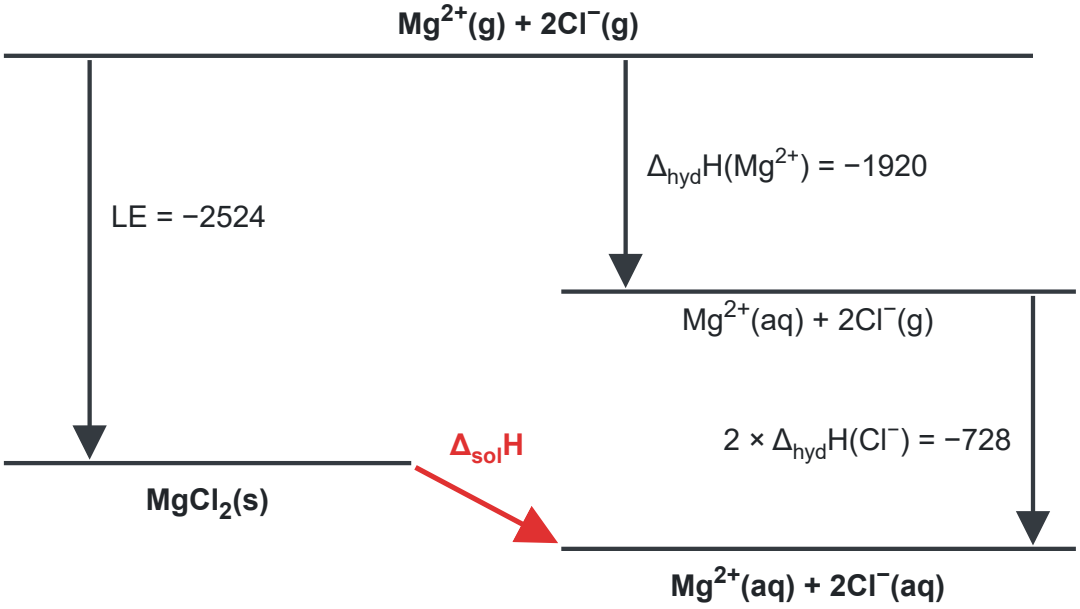
The solution cycle

Dissolving can be imagined in two stages: break the lattice into gaseous ions (the reverse of lattice formation, so +|LE|), then hydrate the ions. Hess's law gives:

Enthalpy change of solution

$$\Delta_{\text{sol}}H = -\text{LE} + \sum \Delta_{\text{hyd}}H$$

LE: lattice energy (negative, for formation) $\sum \Delta_{\text{hyd}}H$: hydration enthalpies of all the ions



Solution cycle for magnesium chloride, values in kJ mol^{-1} (not to scale). The direct route (red) equals the route up through the gaseous ions and down through hydration.

Enthalpy change of solution of magnesium chloride

1. Data / kJ mol^{-1}: LE($MgCl_2$) = −2524; $\Delta_{hyd}H(Mg^{2+})$ = −1920; $\Delta_{hyd}H(Cl^-)$ = −364.
2. Breaking up the lattice is the reverse of lattice formation: +2524 kJ mol^{-1}.
3. Hydrate one Mg^{2+} ion and **two** Cl^- ions:

$$\sum \Delta_{\text{hyd}}H = -1920 + 2(-364) = -2648\ \text{kJ mol}^{-1}$$

4. Add the two stages:

$$\Delta_{\text{sol}}H = +2524 + (-2648) = -124\ \text{kJ mol}^{-1}$$

Exothermic: the hydration of the ions releases more energy than breaking the lattice takes in, and the solution warms up as magnesium chloride dissolves.

Lattice energy and total hydration enthalpy are usually large and similar in size, so $\Delta_{sol}H$, the small difference between them, can come out either side of zero. A very endothermic $\Delta_{sol}H$, where the lattice is far stronger than the hydration can repay, usually means the compound is insoluble. But enthalpy is not the whole story: ammonium

nitrate dissolves readily even though $\Delta_{sol}H$ is +26 kJ mol^{-1} and the solution goes cold. To see why, we need entropy.

Entropy

A **spontaneous** (feasible) change is one that happens on its own once started, without anything continuing to drive it: a gas spreading through a room, ice melting on a warm day, an acid neutralising an alkali. Most feasible reactions are exothermic, but not all. Ice melting and ammonium nitrate dissolving both take in heat. The quantity that decides the direction of every change is **entropy**, S.

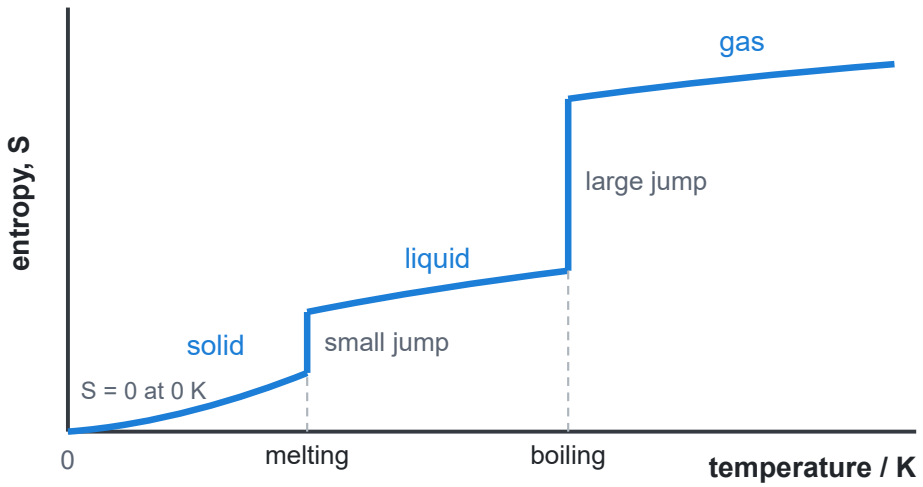
DEFINITION

Entropy is a measure of the number of ways that particles, and the quanta of energy they hold, can be arranged. The more ways there are, the greater the entropy, and the more disordered the system appears.

Energy comes in small packets (quanta) that particles can swap among themselves. With more quanta, or more particles to share them, the number of possible arrangements grows enormously. A system left to itself drifts into the most probable state: the one with the most arrangements, which is why energy spreads out and never concentrates by itself.

Entropy of substances

- **Gases** have far higher entropies than liquids, and liquids higher than solids, because their particles are free to move and can be arranged in vastly more ways.
- Entropy **increases with temperature**: more energy quanta means more ways to arrange them, and particles vibrate and move more.
- There is a sudden jump at each change of state. The jump at boiling is much bigger than at melting, because a gas is so much more disordered than a liquid.
- More complex molecules, with more atoms that can move in more ways, have higher entropies than simple ones; mixtures have higher entropies than pure substances.
- Only a **perfect crystal at 0 K** has zero entropy. Every substance at 298 K, elements included, has a positive standard entropy, $S^{\ominus}$, measured in **J K^{-1} mol^{-1}** (note: joules, not kilojoules).



Entropy rises with temperature, with a small jump on melting and a much larger one on boiling.

Predicting the sign of ΔS_{system}

The **system** is the reacting chemicals; everything else (the flask, the water, the air) is the **surroundings**. You can usually predict the sign of the system's entropy change from the equation. ΔS_{system} is positive when a solid or liquid turns into a gas, and when the number of moles of gas increases; it is negative when the reverse happens.

- $NH_4Cl(s) \rightarrow NH_3(g) + HCl(g)$: a solid becomes two gases, so ΔS_{system} is large and positive.
- $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$: a gas is used up to make a solid, so ΔS_{system} is negative.
- $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$: four moles of gas become two, so ΔS_{system} is negative.
- $NaCl(s) + aq \rightarrow Na^+(aq) + Cl^-(aq)$: an ordered lattice breaks into freely moving ions, so ΔS_{system} is usually positive.

Calculating ΔS_{system}

Entropy change of the system

$$\Delta S^\circ_{\text{system}} = \sum S^\circ_{\text{products}} - \sum S^\circ_{\text{reactants}}$$

Multiply each standard entropy by its number of moles in the equation Units: J K^{-1} mol^{-1}

ΔS for $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

1. Data / J K^{-1} mol^{-1}: $S^\ominus[N_2(g)] = 191.6$; $S^\ominus[H_2(g)] = 130.6$; $S^\ominus[NH_3(g)] = 192.3$.

2. Products: 2 × 192.3 = 384.6. Reactants: 191.6 + 3 × 130.6 = 583.4.

3. $$\Delta S_{\text{system}} = 384.6 - 583.4 = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$$

Negative, as predicted: four moles of gas become two.

Entropy of the surroundings and total entropy

An exothermic reaction releases energy into the surroundings. Those extra quanta can be shared among the surrounding particles in many more ways, so the entropy of the surroundings **increases**. An endothermic reaction takes energy from the surroundings and lowers their entropy. The effect is bigger when the surroundings are cold: adding a given amount of energy to something cold makes a larger relative difference to the number of ways it can be shared than adding it to something already hot.

Entropy change of the surroundings and total entropy change

$$\Delta S_{\text{surroundings}} = -\frac{\Delta H}{T} \qquad \Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

ΔH in J mol⁻¹ (× 1000 from kJ) T in K Entropies in J K⁻¹ mol⁻¹

KEY IDEA

A change is **feasible** (thermodynamically possible) if **ΔS_{total} is positive**. This is the second law of thermodynamics: the total entropy of the system and its surroundings increases in every spontaneous change.

Is decomposing calcium carbonate feasible at 298 K?

1. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$. Data: $\Delta_f H$ / kJ mol^{-1}: $CaCO_3$ −1206.9, CaO −635.1, CO_2 −393.5. $S^{\ominus}$ / J K^{-1} mol^{-1}: $CaCO_3$ 92.9, CaO 39.7, CO_2 213.6.

2. Enthalpy change: $\Delta H = (-635.1 - 393.5) - (-1206.9) = +178.3$ kJ mol^{-1}.

3. Entropy change of the system: $\Delta S_{system} = (39.7 + 213.6) - 92.9 = +160.4$ J K^{-1} mol^{-1}.

Positive, because a gas is made from a solid.

4. Entropy change of the surroundings at 298 K, converting ΔH to J mol^{-1}:

$$\Delta S_{\text{surroundings}} = -\frac{178\,300}{298} = -598.3\ \text{J K}^{-1}\,\text{mol}^{-1}$$

5. Total:

$$\Delta S_{\text{total}} = +160.4 + (-598.3) = -437.9\ \text{J K}^{-1}\,\text{mol}^{-1}$$

Negative, so limestone does not decompose at room temperature.

6. At what temperature does it become feasible? At the boundary $\Delta S_{\text{total}} = 0$, so $\Delta S_{\text{system}} = \Delta H/T$:

$$T = \frac{\Delta H}{\Delta S_{\text{system}}} = \frac{178\,300}{160.4} = 1110\ \text{K}$$

Above about 1110 K (840 °C), $\Delta S_{\text{surroundings}}$ is small enough for the positive ΔS_{system} to win. Lime kilns run hotter than this.

Changes of state

At its melting or boiling temperature, a substance is in equilibrium between the two states, and $\Delta S_{\text{total}} = 0$. That makes $T = \Delta H/\Delta S_{\text{system}}$ a way to calculate the temperature of a change of state. For water boiling, ΔH = +40.7 kJ mol^{-1} and ΔS_{system} = +109 J K^{-1} mol^{-1}, so T = 40 700 / 109 = 373 K. Below 373 K, $\Delta S_{\text{surroundings}}$ is more negative than ΔS_{system} is positive, and liquid water is the stable state; above it, the reverse.

How temperature affects feasibility

Since $\Delta S_{\text{total}} = \Delta S_{\text{system}} - \Delta H/T$, raising the temperature shrinks the $-\Delta H/T$ term. Whether that helps depends on the signs:

ΔH	ΔS_{system}	Feasible?	Example
− (exothermic)	+	at every temperature	$2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$
− (exothermic)	−	only **below** T = ΔH/ΔS	$N_2 + 3H_2 \rightarrow 2NH_3$, below about 464 K
+ (endothermic)	+	only **above** T = ΔH/ΔS	$CaCO_3 \rightarrow CaO + CO_2$, above about 1110 K
+ (endothermic)	−	never	$N_2(g) + 2O_2(g) \rightarrow 2NO_2(g)$

If ΔH is close to zero, −ΔH/T is small at any temperature, and feasibility depends almost entirely on ΔS_{system}.

Gibbs energy

Chemists usually combine ΔH and ΔS_{system} into one quantity that refers only to the system: the **Gibbs energy change**, ΔG.

From ΔS_total to ΔG

1. Start from the total entropy change:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} - \frac{\Delta H}{T}$$

2. Multiply every term by −T:

$$-T\Delta S_{\text{total}} = \Delta H - T\Delta S_{\text{system}}$$

3. Define the right-hand side as the Gibbs energy change:

$$\Delta G = \Delta H - T\Delta S_{\text{system}} = -T\Delta S_{\text{total}}$$

Because T is always positive, ΔG is negative exactly when ΔS_{total} is positive.

Gibbs energy change

$$\Delta G = \Delta H - T\Delta S_{\text{system}}$$

ΔG, ΔH in kJ mol⁻¹ ΔS in kJ K⁻¹ mol⁻¹ (÷ 1000 from J) T in K

KEY IDEA

A reaction is **feasible when ΔG is negative**. When ΔG = 0 the system is at the boundary, in equilibrium. A graph of ΔG against T is a straight line with intercept ΔH and gradient $-\Delta S_{system}$, and it crosses zero at T = $\Delta H/\Delta S_{system}$.

ΔG for decomposing calcium carbonate

1. $\Delta H = +178.3$ kJ mol^{-1} and $\Delta S_{system} = +160.4$ J K^{-1} mol^{-1} = +0.1604 kJ K^{-1} mol^{-1}.

2. At 298 K:

$$\Delta G = 178.3 - 298 \times 0.1604 = +130.5 \text{ kJ mol}^{-1}$$

Positive: not feasible, as the entropy calculation showed.

3. At 1200 K:

$$\Delta G = 178.3 - 1200 \times 0.1604 = -14.2 \text{ kJ mol}^{-1}$$

Negative: feasible at kiln temperatures.

Gibbs energy and equilibrium constants

ΔG is linked to the equilibrium constant K (K_c or K_p) of the reaction by

Gibbs energy and the equilibrium constant

$$\Delta G = -RT \ln K$$

R = 8.31 J K⁻¹ mol⁻¹, so ΔG here is in J mol⁻¹ T in K

A negative ΔG means ln K is positive, so K > 1 and products predominate at equilibrium. The more negative ΔG, the larger K. A reaction with ΔG of about −60 kJ mol^{-1} at 298 K has K of about 10^{10}: it goes essentially to completion. A positive ΔG gives K < 1, but not zero, so a little product still forms.

K	below 10^{-10}	about 0.1	1	about 10	above 10^{10}
Position of equilibrium	reaction effectively doesn't go	reactants predominate	similar amounts	products predominate	reaction effectively complete

Feasible does not mean fast

ΔG and ΔS_{total} say whether a reaction *can* happen, not how quickly. A mixture of hydrogen and oxygen at room temperature has a hugely negative ΔG for forming water, yet nothing happens until a spark supplies the activation energy. Diamond should turn into graphite at room temperature (ΔG is negative), but the rate is immeasurably slow. Such systems are **thermodynamically unstable but kinetically stable** (kinetically inert): a high activation energy

stops the feasible reaction from happening at a noticeable rate. Remember too that the calculations use standard data and assume ΔH and ΔS stay the same at other temperatures, which is only approximately true.

Entropy and solubility

Now return to dissolving. When an ionic solid dissolves, the ions leave an ordered lattice and spread through the solution, so ΔS_{system} is usually positive.

- If $\Delta_{sol}H$ is **negative**, $\Delta S_{surroundings}$ is positive too, so ΔS_{total} is positive and the solid dissolves at any temperature.
- If $\Delta_{sol}H$ is **positive**, $\Delta S_{surroundings}$ is negative. The solid still dissolves if the positive ΔS_{system} is bigger. Ammonium nitrate: $\Delta_{sol}H$ = +25.7 kJ mol^{-1}, ΔS_{system} = +108.7 J K^{-1} mol^{-1}. At 298 K, $\Delta S_{surroundings}$ = −25 700/298 = −86.2, so ΔS_{total} = +22.5 J K^{-1} mol^{-1}: it dissolves, and cold packs use exactly this.
- Raising the temperature makes the negative $-\Delta_{sol}H/T$ term smaller in size, so salts with an endothermic $\Delta_{sol}H$ generally become more soluble when heated.
- If $\Delta_{sol}H$ is very endothermic, because the lattice energy is far larger than the hydration enthalpies, $\Delta S_{surroundings}$ is too negative to overcome and the solid is insoluble.

Quick checks

QUICK CHECK

Which equation represents the lattice energy of magnesium oxide?

- $Mg(s) + ½O_2(g) \rightarrow MgO(s)$
- $MgO(s) \rightarrow Mg^{2+}(g) + O^{2-}(g)$
- ✓ $Mg^{2+}(g) + O^{2-}(g) \rightarrow MgO(s)$
- $Mg^{2+}(aq) + O^{2-}(aq) \rightarrow MgO(s)$

Lattice energy is the formation of 1 mol of solid from its gaseous ions. The first option is the enthalpy of formation; the second is the lattice breaking up (same size, opposite sign); aqueous ions are wrong because hydration energy would be involved.

QUICK CHECK

Use the data to find the first electron affinity of bromine. Data / kJ mol^{-1}: $\Delta_f H$(KBr) = −394; $\Delta_{at} H$(K) = +89; first IE(K) = +419; $\Delta_{at} H$(Br) = +112; LE(KBr) = −689.

- +325 kJ mol^{-1}
- −1703 kJ mol^{-1}
- −915 kJ mol^{-1}
- −325 kJ mol^{-1}

$\Delta_f H = \Delta_{at} H(K) + IE + \Delta_{at} H(Br) + EA + LE$, so EA = −394 − (89 + 419 + 112 − 689) = −394 − (−69) = −325 kJ mol^{-1}. A positive answer comes from a sign error with the lattice energy.

QUICK CHECK

The Born–Haber lattice energy of a compound is much more exothermic than its theoretical lattice energy. What does this show?

- The ions are larger than expected.
- The bonding has significant covalent character.
- The compound is purely ionic.
- The compound will be very soluble in water.

The theoretical value assumes perfectly spherical ions. A polarising cation distorts the anion, putting electron density between the ions; this covalent character strengthens the lattice beyond what the ionic model predicts.

QUICK CHECK

Which change has the largest positive ΔS_{system}?

- $H_2O(g) \rightarrow H_2O(l)$
- $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
- $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ ✓
- $Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$

Only the decomposition of calcium carbonate creates a gas from a solid. The other three all reduce the amount or freedom of gas particles, so their ΔS_{system} is negative.

QUICK CHECK

A reaction has $\Delta H = -120$ kJ mol^{-1} and $\Delta S_{system} = -150$ J K^{-1} mol^{-1}. Above what temperature is it no longer feasible?

- 0.8 K
- 800 K ✓
- 1250 K
- It is feasible at all temperatures.

Feasible while $\Delta G = \Delta H - T\Delta S$ is negative. $\Delta G = 0$ when $T = \Delta H/\Delta S = -120\,000 / -150 = 800$ K. Above that, $-T\Delta S$ (positive) outweighs the negative ΔH. Forgetting to convert kJ to J gives 0.8 K.

QUICK CHECK

Why is the hydration enthalpy of Mg^{2+} much more exothermic than that of Na^{+}?

- Mg^{2+} has more electrons than Na^{+}.
- Mg^{2+} forms covalent bonds with water.
- Mg^{2+} is smaller and has a larger charge, so it attracts the δ− oxygen of water molecules more strongly.
- Mg^{2+} is surrounded by hydrogen atoms of water molecules.

Hydration energy comes from ion–dipole attraction. Mg^{2+} has a much higher charge density than Na^{+} (twice the charge, smaller radius), so water's δ− oxygen atoms are pulled in closer and more strongly. Cations are surrounded by oxygen, not hydrogen, atoms.

QUICK CHECK

The conversion of diamond into graphite has a negative ΔG at room temperature, yet diamonds do not change. Why?

- ΔS_{total} for the change is negative.
- The activation energy is so high that the rate is negligible.
- The equilibrium constant is less than 1.
- Graphite has a lower entropy than diamond.

A negative ΔG means ΔS_{total} is positive and K > 1, so the change is feasible. Feasibility says nothing about rate: diamond is kinetically stable because the activation energy for rearranging its strong covalent network is enormous.

A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Chemical equilibria

At AS you predicted which way an equilibrium moves. Now you put numbers on it. The equilibrium constants K_c and K_p measure exactly where an equilibrium lies, you can calculate them from experimental data, and they explain why changing the temperature behaves so differently from changing concentration, pressure or adding a catalyst.

The equilibrium constant K_c

In a closed system a reversible reaction settles into **dynamic equilibrium**: the forward and backward reactions are still happening, but at the same rate, so the concentrations of everything stop changing. For a given reaction at a given temperature, the equilibrium concentrations always fit one fixed ratio, whatever amounts you started with. That ratio is the equilibrium constant.

For the general reaction $mA + nB \rightleftharpoons pC + qD$:

$$K_c = \frac{[C]^p\,[D]^q}{[A]^m\,[B]^n}$$

- **Square brackets** mean the concentration at equilibrium, in mol dm^{-3}.
- **Products go on top**, reactants underneath, and each concentration is raised to the power of its number in the balanced equation.
- **K_c belongs to an equation as written.** Write the equation backwards and the new K_c is the reciprocal of the old one; double every number in the equation and K_c is squared. Always quote the equation with the value.

Homogeneous and heterogeneous equilibria

In a **homogeneous** equilibrium everything is in the same phase, for example all gases, or all dissolved in one solution. In a **heterogeneous** equilibrium there is more than one phase, such as a solid in contact with a gas or a solution. A pure solid (or a pure liquid that forms its own separate layer) has a fixed concentration however much of it is there, so it is left out of the K_c expression.

Equilibrium	Type	K_c
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	homogeneous	$\frac{[HI]^2}{[H_2][I_2]}$
$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$	homogeneous (one liquid mixture)	$\frac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]}$

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	heterogeneous	$[CO_2]$
$Ag^+(aq) + Fe^{2+}(aq) \rightleftharpoons Ag(s) + Fe^{3+}(aq)$	heterogeneous	$\dfrac{[Fe^{3+}]}{[Ag^+][Fe^{2+}]}$

COMMON MISTAKE

Water is left out only when it is the solvent, present in huge excess. In the esterification above, water is one of four liquids mixed in a single solution and is formed in similar amounts to the ester, so it **does** appear in K_c.

Working out the units of K_c

K_c has units only when the powers on the top and bottom do not cancel. Put mol dm^{-3} in for every concentration and simplify.

Units of K_c for N₂(g) + 3H₂(g) ⇌ 2NH₃(g)

1. Write the expression: $K_c = \dfrac{[NH_3]^2}{[N_2][H_2]^3}$.
2. Substitute the unit: $\dfrac{(\text{mol dm}^{-3})^2}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})^3} = \dfrac{(\text{mol dm}^{-3})^2}{(\text{mol dm}^{-3})^4}$.
3. Cancel: $(\text{mol dm}^{-3})^{-2} = \text{mol}^{-2}\,\text{dm}^6$.

A quick check: there are 2 molecules on top and 4 underneath, so the unit is (mol dm^{-3})$^{2-4}$.

For $H_2 + I_2 \rightleftharpoons 2HI$ there are 2 on top and 2 underneath, so K_c has **no units**. For $2SO_2 + O_2 \rightleftharpoons 2SO_3$ it is (mol dm^{-3})$^{2-3}$ = dm^3 mol^{-1}.

Calculating K_c from experimental data

A question usually gives the starting amounts of the reactants and one piece of information about the equilibrium mixture: the amount of one substance, or the percentage of a reactant that has reacted. The balanced equation then tells you how everything else changed. Lay it out in a table with three rows, often called an ICE table: **I**nitial moles, **C**hange, **E**quilibrium moles. Then divide by the volume to get concentrations.

- **Reactant at equilibrium** = moles at the start − moles that reacted.
- **Product at equilibrium** = moles at the start + moles that formed.
- The changes are always in the ratio of the numbers in the balanced equation.

Example 1: H₂ + I₂ ⇌ 2HI (no units)

1. 0.500 mol of H_2 and 0.500 mol of I_2 are heated in a sealed 2.00 dm^3 flask. At equilibrium the flask contains 0.786 mol of HI. Find K_c.

2. HI formed = 0.786 mol. The equation says 2 HI form for every 1 H_2 and 1 I_2 used, so 0.393 mol of each reactant reacted.

3.

	H_2	I_2	HI
Initial / mol	0.500	0.500	0
Change / mol	−0.393	−0.393	+0.786
Equilibrium / mol	0.107	0.107	0.786
Concentration / mol dm^{-3}	0.0535	0.0535	0.393

4. $K_c = \dfrac{[\mathrm{HI}]^2}{[\mathrm{H_2}][\mathrm{I_2}]} = \dfrac{0.393^2}{0.0535 \times 0.0535} = 54.0$, with no units.

Because the units cancel, the volumes cancel too: putting the equilibrium moles straight in gives the same answer. That shortcut only works when K_c has no units.

Example 2: 2SO₂ + O₂ ⇌ 2SO₃ (with units)

1. 2.00 mol of SO_2 and 1.20 mol of O_2 are mixed in a 5.00 dm^3 vessel with a catalyst. At equilibrium there are 1.60 mol of SO_3. Find K_c.

2. 1.60 mol of SO_3 formed, so 1.60 mol of SO_2 and 0.800 mol of O_2 reacted (ratio 2 : 1 : 2).

3. Equilibrium moles: SO_2 2.00 − 1.60 = 0.400; O_2 1.20 − 0.800 = 0.400; SO_3 1.60.

4. Divide by 5.00 dm^3: $[SO_2]$ = 0.0800, $[O_2]$ = 0.0800, $[SO_3]$ = 0.320 mol dm^{-3}.

5 $$K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]} = \frac{0.320^2}{0.0800^2 \times 0.0800} = 200\ \text{dm}^3\,\text{mol}^{-1}$$

Here the volume does not cancel. Forgetting to divide by 5.00 would give 40.0, a wrong answer.

Practical: finding K_c for an esterification

Ethanoic acid and ethanol react slowly to make ethyl ethanoate and water. A little strong acid speeds the reaction up so that equilibrium is reached in about a week at room temperature:

$$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$$

Method

1. Use burettes to measure known volumes of ethanoic acid, ethanol and a dilute acid catalyst into a boiling tube. Stopper it, swirl and leave it for a week.
2. Find how much acid the catalyst alone contributes by titrating a separate sample of the same volume of catalyst with sodium hydroxide.
3. Wash the equilibrium mixture into a 250 cm^3 volumetric flask with distilled water, make it up to the mark and invert it to mix.
4. Pipette 25.0 cm^3 portions into conical flasks and titrate with standard sodium hydroxide using phenolphthalein, until you have concordant titres.

- **Starting amounts** of the acid and the alcohol come from their volumes: mass = density × volume, then moles = mass ÷ M_r.
- **The titration measures all the acid present**: the ethanoic acid left at equilibrium plus the catalyst. Subtract the catalyst's share to find the ethanoic acid.
- **Diluting with water does not spoil the result** because the titration is done quickly, and the uncatalysed reaction is far too slow to shift much in that time.
- **The pink end-point can fade**: as the hydroxide removes ethanoic acid, the equilibrium slowly shifts left, hydrolysing some ester and releasing more acid. Take the first permanent pink.
- **To check that a week was long enough**, set up identical mixtures and analyse them after different times. If K_c stops changing, equilibrium had been reached.

Calculating K_c from the titration

1 The mixture started with 0.200 mol of ethanoic acid and 0.200 mol of ethanol. Each 25.0 cm^3 portion of the diluted mixture needed 21.40 cm^3 of 0.500 mol dm^{-3} NaOH. The catalyst alone contains 0.0400 mol of H^+.

2. NaOH used per portion = 0.02140 dm^3 × 0.500 mol dm^{-3} = 0.01070 mol, so the 25.0 cm^3 portion held 0.01070 mol of acid.

3. The whole 250 cm^3 held ten times as much: 0.1070 mol of acid.

4. Ethanoic acid at equilibrium = 0.1070 − 0.0400 = 0.0670 mol.

5. Ethanoic acid that reacted = 0.200 − 0.0670 = 0.133 mol, so 0.133 mol of ethanol reacted, and 0.133 mol each of ester and water formed. Ethanol left = 0.200 − 0.133 = 0.0670 mol.

6. $K_c = \dfrac{0.133 \times 0.133}{0.0670 \times 0.0670} = 3.94$, with no units, so the total volume cancels.

   This ignores the water that came in with the dilute catalyst. Including it makes [H_2O] slightly bigger and K_c slightly bigger: a known limitation of the method.

Partial pressures and K_p

For reactions between gases it is often easier to measure pressures than concentrations, so a second equilibrium constant, K_p, is written with **partial pressures**.

- **Partial pressure** of a gas in a mixture: the pressure it would exert if it alone filled the whole volume of the mixture. It is written p_A or p(A).
- **Mole fraction**: $x_A = \dfrac{\text{moles of A}}{\text{total moles of gas}}$. The mole fractions of all the gases add up to 1.
- **Partial pressure** = mole fraction × total pressure: $p_A = x_A P$.
- **The partial pressures add up to the total pressure**: $P = p_A + p_B + p_C + \ldots$

For $mA(g) + nB(g) \rightleftharpoons pC(g) + qD(g)$:

$$K_p = \frac{p_C^{\,p}\, p_D^{\,q}}{p_A^{\,m}\, p_B^{\,n}}$$

Only gases appear in K_p. Solids, liquids and dissolved substances are left out, so for $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $K_p = p_{CO_2}$, with units of pressure. Units of K_p are worked out like those of K_c, using the pressure unit given in the question (atm or kPa): for $N_2 + 3H_2 \rightleftharpoons 2NH_3$, K_p is in atm^{-2} (or kPa^{-2}).

Example 3: N₂O₄(g) ⇌ 2NO₂(g)

1. 1.00 mol of N_2O_4 is allowed to reach equilibrium. 20.0% of it dissociates, and the total pressure is 1.00 atm. Find K_p.

2. N_2O_4 reacted = 0.200 mol, so NO_2 formed = 0.400 mol. At equilibrium: 0.800 mol N_2O_4 and 0.400 mol NO_2, 1.200 mol in total.

3. Mole fractions: $x_{N_2O_4} = 0.800/1.200 = \frac{2}{3}$; $x_{NO_2} = 0.400/1.200 = \frac{1}{3}$.

4. Partial pressures at 1.00 atm: $p_{N_2O_4} = \frac{2}{3} = 0.667$ atm, $p_{NO_2} = \frac{1}{3} = 0.333$ atm.

5. $$K_p = \frac{p_{NO_2}^2}{p_{N_2O_4}} = \frac{(1/3)^2}{2/3} = 0.167 \text{ atm}$$

   With the rounded values 0.333 and 0.667 you would get 0.166: another reason to keep full figures until the end.

Example 4: N₂(g) + 3H₂(g) ⇌ 2NH₃(g)

1. 2.00 mol of N_2 and 6.00 mol of H_2 reach equilibrium at a total pressure of 100 atm. The equilibrium mixture contains 1.00 mol of NH_3. Find K_p.

2. 1.00 mol of NH_3 formed, so 0.500 mol of N_2 and 1.50 mol of H_2 reacted. At equilibrium: N_2 1.50, H_2 4.50, NH_3 1.00; total 7.00 mol.

3. Partial pressures: $p_{N_2} = \frac{1.50}{7.00} \times 100 = 21.4$ atm; $p_{H_2} = \frac{4.50}{7.00} \times 100 = 64.3$ atm; $p_{NH_3} = \frac{1.00}{7.00} \times 100 = 14.3$ atm. Check: they add up to 100 atm.

4. $$K_p = \frac{p_{NH_3}^2}{p_{N_2}\, p_{H_2}^3} = \frac{14.3^2}{21.4 \times 64.3^3} = 3.58 \times 10^{-5} \text{ atm}^{-2}$$

   Keep the unrounded partial pressures in your calculator; rounding to three figures first can shift the last digit.

What changes K, and what does not

KEY IDEA

For a given equation, the value of K_c or K_p depends **only on temperature**. Changing concentration or pressure moves the **position** of equilibrium but leaves K unchanged, and a catalyst changes neither.

Concentration

Add more H_2 to an equilibrium mixture of H_2, I_2 and HI. For a moment the fraction $\frac{[HI]^2}{[H_2][I_2]}$ is smaller than K_c, because its bottom line has grown. The system is no longer at equilibrium. The forward reaction now runs faster than the backward one, using up H_2 and I_2 and making HI, until the fraction is back up to exactly K_c. The position has moved right; K_c has not changed.

Pressure

Increasing the total pressure changes every partial pressure at once, but not by the same overall factor on the top and bottom of K_p when the two sides have different numbers of gas molecules.

Why higher pressure gives more ammonia although K_p stays the same

1. Write each partial pressure as mole fraction × total pressure: $K_p = \dfrac{(x_{NH_3}P)^2}{(x_{N_2}P)(x_{H_2}P)^3}$.
2. Collect the P terms: $K_p = \dfrac{x_{NH_3}^2}{x_{N_2}\,x_{H_2}^3} \times \dfrac{1}{P^2}$.
3. Double P and the factor $1/P^2$ falls to a quarter. Straight after the change the right-hand side is smaller than K_p, so the mixture is not at equilibrium.
4. To restore K_p, the mole-fraction part must grow fourfold: more NH_3 forms and N_2 and H_2 are used up. The position shifts right, towards fewer gas molecules, and K_p ends where it started.

If both sides have the same number of gas molecules, the P terms cancel completely and pressure has no effect on the position at all.

Catalysts

A catalyst provides an alternative route with a lower activation energy for both the forward and the backward reaction. Both speed up by the same factor, so equilibrium is reached sooner but its position and the value of K are unchanged.

Temperature

Temperature is the one condition that changes K itself. If the forward reaction is **exothermic**, raising the temperature makes K **smaller**: the equilibrium mixture contains less product. If the forward reaction is **endothermic**, raising the temperature makes K **larger**. The new value of K then forces the concentrations to adjust, which is why the position also moves. The next section explains why, using entropy.

The size of K, extent of reaction and entropy

The bigger K is, the further the reaction goes before it reaches equilibrium. As a rough guide:

Value of K	What the equilibrium mixture is like
less than about 10^{-10}	essentially only reactants: the reaction does not happen to any noticeable extent
around 0.1	reactants predominate
around 1	similar amounts of reactants and products
around 10	products predominate
more than about 10^{10}	essentially only products: the reaction goes to completion

This links to Topic 12. The total entropy change of a reaction and its equilibrium constant are connected by

$$\Delta S_{\text{total}} = R \ln K$$

where R = 8.31 J K^{-1} mol^{-1}. A positive ΔS_{total} means K > 1, so products are favoured; a negative ΔS_{total} means K < 1. A reaction with K = 10^{10} has ΔS_{total} = 8.31 × ln 10^{10} = +191 J K^{-1} mol^{-1}. Since $\Delta G = -T\Delta S_{total}$, the same idea can be written $\Delta G = -RT \ln K$.

Why temperature changes K

1. $\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$, and $\Delta S_{\text{surroundings}} = -\frac{\Delta H}{T}$.
2. For an **exothermic** reaction ΔH is negative, so $\Delta S_{surroundings}$ is positive. As T rises, −ΔH/T gets **smaller**.
3. So ΔS_{total} becomes less positive (or more negative). Because ΔS_{total} = R ln K, ln K falls and K gets smaller: less product at equilibrium.
4. For an **endothermic** reaction $\Delta S_{surroundings}$ is negative, and rising T makes it less negative. ΔS_{total} increases, so K gets larger.

This is the quantitative reason behind the Le Chatelier rule you learned at AS: heating favours the endothermic direction.

Equilibria in industry: yield against rate

An industrial chemist wants a high yield *and* a fast rate at a sensible cost. Equilibrium and kinetics often pull in opposite directions, so the conditions chosen are a compromise.

	Haber process $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta H = -92$ kJ mol^{-1}	**Contact process** $2SO_2 + O_2 \rightleftharpoons 2SO_3$, $\Delta H = -197$ kJ mol^{-1}
Temperature	about 450 °C. Lower would give a bigger K and more ammonia, but the rate would be far too slow.	about 450 °C, for the same reason.
Pressure	about 200 atm. Higher pressure favours the side with fewer gas molecules (4 → 2) and speeds the reaction, but costs more to build and run safely.	1 to 2 atm. K is already so large that the yield is about 99% without high pressure, so the expense is not worth it.
Catalyst	iron: equilibrium reached faster, K unchanged.	vanadium(V) oxide, V_2O_5.
Other	ammonia is liquefied and removed, and unreacted gases are recycled.	SO_3 is removed by absorbing it, and unreacted gases pass over further catalyst beds.

EXAM TIP

When explaining a choice of conditions, give both sides: what the condition does to the **equilibrium yield** (through K or the position) and what it does to the **rate** and the **cost**. A catalyst helps the rate and lets a lower temperature be used, saving energy, but never improves the yield.

Quick checks

QUICK CHECK

What are the units of K_c for $2NO_2(g) \rightleftharpoons N_2O_4(g)$?

- mol dm^{-3}
- dm^3 mol^{-1}
- no units
- mol^2 dm^{-6}

$K_c = [N_2O_4]/[NO_2]^2$: one concentration on top and two underneath gives (mol dm^{-3})$^{-1}$ = dm^3 mol^{-1}. mol dm^{-3} is the unit for the reverse equation.

QUICK CHECK

Which is the correct K_p expression for $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$?

- $\dfrac{p_{CO}\, p_{H_2}}{p_C\, p_{H_2O}}$
- $\dfrac{p_{H_2O}}{p_{CO}\, p_{H_2}}$
- $\dfrac{p_{CO}\, p_{H_2}}{p_{H_2O}}$
- $p_{CO}\, p_{H_2}$

Carbon is a solid, so it does not appear; steam is a gas and does. Products go on top.

QUICK CHECK

A gas mixture contains 0.60 mol of A and 1.40 mol of B at a total pressure of 250 kPa. What is the partial pressure of A?

- 75 kPa
- 107 kPa
- 150 kPa
- 0.30 kPa

$x_A = 0.60 \div 2.00 = 0.30$, so $p_A = 0.30 \times 250 = 75$ kPa. 107 kPa comes from dividing by the moles of B instead of the total.

QUICK CHECK

For an exothermic equilibrium, what happens when the temperature is raised?

- K increases and the position moves right
- K is unchanged but the position moves left
- K decreases but the position does not change
- K decreases and the position moves left

Temperature is the only thing that changes K. For an exothermic reaction $\Delta S_{surroundings} = -\Delta H/T$ falls as T rises, so ΔS_{total} and K fall, and the mixture holds less product.

QUICK CHECK

The pressure on the equilibrium $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is increased at constant temperature. What happens to K_p and to the amount of SO_3?

- K_p increases; more SO_3
- K_p unchanged; more SO_3
- K_p unchanged; less SO_3
- K_p decreases; more SO_3

K_p depends only on temperature. With 3 gas molecules on the left and 2 on the right, higher pressure shifts the position right, so the amount of SO_3 rises.

QUICK CHECK

A reaction has $K_c = 2 \times 10^{12}$ at 298 K. What can you say about it?

- It is fast
- Reactants predominate at equilibrium
- It goes essentially to completion, and ΔS_{total} is positive
- ΔS_{total} is negative

K above about 10^{10} means essentially complete reaction, and $\Delta S_{total} = R \ln K$ is positive. K says nothing about the rate: a reaction can have a huge K and still be very slow.

QUICK CHECK

0.40 mol of PCl_5 is heated in a 2.0 dm^3 flask: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. At equilibrium 0.10 mol of Cl_2 is present. What is K_c?

- 0.033 mol dm^{-3}
- 0.067 mol dm^{-3}
- 0.025 mol dm^{-3}
- 0.017 mol dm^{-3}

Equilibrium: PCl_5 0.30 mol, PCl_3 0.10 mol, Cl_2 0.10 mol. Concentrations 0.15, 0.050 and 0.050 mol dm^{-3}, so $K_c = (0.050 \times 0.050) \div 0.15 = 0.017$ mol dm^{-3}. Using moles without dividing by the volume gives 0.033.

QUICK CHECK

Why is the Haber process run at about 450 °C rather than at a lower temperature?

- K is larger at 450 °C
- A lower temperature would give a better yield but far too slow a rate
- The iron catalyst increases K at 450 °C
- Ammonia decomposes below 450 °C

The forward reaction is exothermic, so K is actually larger at lower temperatures. 450 °C is a compromise that gives an acceptable rate with a reasonable yield. A catalyst never changes K.

A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Acid-base equilibria

Acids and bases are all about protons moving from one particle to another. This topic turns that idea into numbers: the pH scale, the ionic product of water, the acid dissociation constant K_a, the shapes of titration curves, and the buffers that keep the pH of your blood almost perfectly steady.

Brønsted–Lowry acids and bases

- **A Brønsted–Lowry acid** is a proton (H^+) donor.
- **A Brønsted–Lowry base** is a proton acceptor. It needs a lone pair of electrons to bond to the proton.

An acid can only give away a proton if something accepts it, so acid–base reactions always involve two pairs. When an acid loses its proton, what is left can accept a proton back: it is the acid's **conjugate base**. Likewise a base that gains a proton becomes its **conjugate acid**. A conjugate pair differs by exactly one H^+.

Reaction	Acid 1	Base 2	Acid 2	Base 1
$HCl + H_2O \rightarrow H_3O^+ + Cl^-$	HCl	H_2O	H_3O^+	Cl^-
$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$	H_2O	NH_3	NH_4^+	OH^-
$HNO_3 + HNO_2 \rightleftharpoons NO_3^- + H_2NO_2^+$	HNO_3	HNO_2	$H_2NO_2^+$	NO_3^-

Water acts as a base towards HCl but as an acid towards NH_3: a substance that can do both is called **amphoteric**. In the last example nitrous acid is forced to act as a base, because nitric acid is the stronger acid; when two acids are mixed, the one with the larger K_a donates the proton.

pH and strong acids

$$\mathrm{pH} = -\log_{10}[\mathrm{H}^+] \qquad [\mathrm{H}^+] = 10^{-\mathrm{pH}}$$

The pH scale is logarithmic so that hydrogen-ion concentrations ranging over many powers of ten fit onto a scale of roughly 0 to 14. Each fall of one pH unit means ten times more H^+. Give pH values to two decimal places.

A **strong acid** is fully dissociated in water, so for a monoprotic strong acid such as HCl or HNO_3, $[H^+]$ equals the concentration of the acid.

pH of a strong acid, and back again

1. 0.0250 mol dm^{-3} nitric acid: [H$^+$] = 0.0250 mol dm^{-3}, so pH = −log 0.0250 = **1.60**.

2. A hydrochloric acid solution has pH 2.45: [H$^+$] = $10^{-2.45}$ = 3.55 × 10^{-3} mol dm^{-3}, and that is also the concentration of the acid.

The ionic product of water, K_w

Water itself ionises very slightly: $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$. So little water ionises that $[H_2O]$ is effectively constant, and it is folded into the constant to give the **ionic product of water**:

$$K_w = [\mathrm{H^+}][\mathrm{OH^-}] = 1.00 \times 10^{-14}\ \mathrm{mol^2\,dm^{-6}}\ \text{at}\ 298\ \mathrm{K} \qquad \mathrm{p}K_w = -\log K_w = 14.00$$

K_w applies to every aqueous solution, so if you know [OH$^-$] you can find [H$^+$], and the other way round.

Neutral water is not always pH 7

A solution is **neutral** when [H$^+$] = [OH$^-$]. At 298 K that gives [H$^+$] = √(1.00 × 10^{-14}) = 1.00 × 10^{-7} mol dm^{-3} and pH 7.00. The ionisation of water is **endothermic**, so warming it increases K_w. At 40 °C, K_w = 2.92 × 10^{-14} mol^2 dm^{-6}, so [H$^+$] = √(2.92 × 10^{-14}) = 1.71 × 10^{-7} mol dm^{-3} and pH = 6.77. That water is still neutral, because [H$^+$] and [OH$^-$] are still equal; neutral simply has a lower pH at higher temperatures.

pH of strong bases

pH of a strong base

1. Strong bases such as NaOH are fully dissociated, so 0.0500 mol dm^{-3} NaOH has [OH$^-$] = 0.0500 mol dm^{-3}.

2. Use K_w: $[\mathrm{H^+}] = \dfrac{K_w}{[\mathrm{OH^-}]} = \dfrac{1.00 \times 10^{-14}}{0.0500} = 2.00 \times 10^{-13}\ \mathrm{mol\,dm^{-3}}$.

3. pH = −log(2.00 × 10^{-13}) = **12.70**.

Watch for bases that release two hydroxide ions: 0.0200 mol dm^{-3} $Ba(OH)_2$ gives [OH$^-$] = 0.0400 mol dm^{-3} and pH 12.60.

Weak acids and K_a

A **weak acid** is only slightly dissociated: the solution is an equilibrium mixture that is mostly undissociated HA. Its strength is measured by the **acid dissociation constant**, K_a, which is simply K_c for the dissociation:

$$\mathrm{HA(aq)} \rightleftharpoons \mathrm{H^+(aq)} + \mathrm{A^-(aq)} \qquad K_a = \frac{[\mathrm{H^+}][\mathrm{A^-}]}{[\mathrm{HA}]} \qquad \mathrm{p}K_a = -\log K_a$$

- The units of K_a are mol dm^{-3}.
- The **larger K_a** (the smaller pK_a), the further the equilibrium lies to the right and the stronger the acid. Ethanoic acid has $K_a = 1.7 \times 10^{-5}$ mol dm^{-3}, pK_a = 4.77.
- **Strong and weak are not the same as concentrated and dilute.** Strength is about how much the acid dissociates; concentration is about how much acid is in each dm^3.

Calculating the pH of a weak acid

Two approximations make the calculation simple:

1. [H$^+$] = [A$^-$], because each HA that dissociates gives one of each. (This ignores the tiny amount of H$^+$ from the water.)
2. [HA] at equilibrium ≈ the concentration of acid you started with, because so little of it dissociates.

Then $K_a = \dfrac{[\mathrm{H^+}]^2}{[\mathrm{HA}]_{\text{initial}}}$, so $[\mathrm{H^+}] = \sqrt{K_a \times [\mathrm{HA}]_{\text{initial}}}$. The second approximation breaks down for acids that are fairly strong (large K_a) or very dilute, because then a significant fraction of the acid dissociates.

pH of 0.0500 mol dm^{−3} ethanoic acid

1. $[\mathrm{H^+}]^2 = K_a \times [\mathrm{HA}] = 1.7 \times 10^{-5} \times 0.0500 = 8.5 \times 10^{-7}$
2. $[\mathrm{H^+}] = \sqrt{8.5 \times 10^{-7}} = 9.22 \times 10^{-4}\ \mathrm{mol\,dm^{-3}}$
3. pH = −log(9.22 × 10^{-4}) = **3.04**.

A strong acid of the same concentration would have pH 1.30. Only about 2% of the ethanoic acid molecules are dissociated.

K_a from a measured pH

1. 0.200 mol dm^{-3} propanoic acid has pH 2.79. Find K_a and pK_a.

2. $[H^+] = 10^{-2.79} = 1.62 \times 10^{-3}$ mol dm^{-3} = $[A^-]$.

3. $K_a = \dfrac{(1.62 \times 10^{-3})^2}{0.200} = 1.32 \times 10^{-5}\ \text{mol dm}^{-3}$, so $pK_a = 4.88$.

Diluting acids

Diluting a strong acid ten times divides $[H^+]$ by ten, so the pH rises by exactly 1. Diluting a weak acid ten times raises its pH by only about 0.5: 0.0500 mol dm^{-3} ethanoic acid has pH 3.04, but 0.00500 mol dm^{-3} has pH 3.54. Adding water shifts the dissociation equilibrium to the right, so a larger fraction of the acid dissociates and partly makes up the lost H^+. Mathematically, $[H^+]$ is proportional to the square root of the concentration.

KEY IDEA

Core practical: finding K_a of a weak acid. Calibrate a pH meter with buffers of known pH. Titrate 25.0 cm^3 of the weak acid with sodium hydroxide to find the volume needed for neutralisation. Then add exactly half that volume of alkali to a fresh 25.0 cm^3 of acid and measure the pH. At half-neutralisation $[HA] = [A^-]$, so pH = pK_a (explained below).

Titration curves

A titration curve plots pH against the volume of one solution added to the other. To record one, put a known volume of acid in a beaker with a calibrated pH meter, add the alkali from a burette a little at a time, stirring and recording the pH after each addition, with smaller additions near the end point. The **equivalence point** is where the acid and base have reacted exactly in the ratio of the equation; on the graph it is the middle of the steep, nearly vertical part.

Titration (base added to 0.1 mol dm^{-3} acid)	Starting pH	Steep section	pH at equivalence
strong acid + strong base (HCl + NaOH)	about 1	about 3 to 11	7
weak acid + strong base (CH_3COOH + NaOH)	about 3	about 7 to 11	above 7 (about 8.7)
strong acid + weak base (HCl + NH_3)	about 1	about 3 to 7	below 7 (about 5.3)
weak acid + weak base (CH_3COOH + NH_3)	about 3	no steep section	about 7, but hard to pin down

The pH at equivalence is not always 7 because the salt formed can itself react with water. Sodium ethanoate contains ethanoate ions, the conjugate base of a weak acid, which take protons from water and leave OH^-: the solution is alkaline. Ammonium chloride contains NH_4^+, a weak acid, so its solution is acidic.

Half-neutralisation: pH = pK_a

On the curve for a weak acid with a strong base, the pH rises quickly at first and then flattens into a **buffer region**. Halfway to the equivalence point, exactly half of the acid has been turned into its salt, so [HA] = [A^-]. These cancel in the K_a expression, leaving $K_a = [H^+]$, so **pH = pK_a**. This is the basis of the core practical above.

Choosing an indicator

An indicator is itself a weak acid, HIn, whose undissociated form and conjugate base have different colours: $HIn \rightleftharpoons H^+ + In^-$. Adding acid pushes the equilibrium left, giving the colour of HIn; adding alkali removes H^+ and pulls it right, giving the colour of In^-. The **end point**, where the colour changes, is when [HIn] = [In^-]; it happens over a range of about 2 pH units centred on the indicator's pK_{in}.

A suitable indicator changes colour completely within the **steep part** of the titration curve, so that one drop of titrant takes it all the way through its colour change at the equivalence point.

Indicator	Colour in acid → alkali	pH range	Use for
methyl orange	red → yellow (orange at the end point)	about 3.1 to 4.4	strong acid with strong or weak base
phenolphthalein	colourless → pink	about 8.3 to 10.0	strong or weak acid with strong base

For a weak acid with a weak base there is no steep section, so no indicator gives a sharp end point; use a pH meter instead.

Diprotic acids such as ethanedioic acid, $H_2C_2O_4$, lose their two protons one at a time, so their curve with sodium hydroxide has two steep sections and two equivalence points. In the same way, adding acid to sodium carbonate gives two steps: $CO_3^{2-} + H^+ \rightarrow HCO_3^-$, then $HCO_3^- + H^+ \rightarrow H_2O + CO_2$.

Buffer solutions

A **buffer** is a solution whose pH changes very little when small amounts of acid or alkali are added, or when it is diluted.

- **Acidic buffer**: a weak acid and a salt of that acid, for example ethanoic acid and sodium ethanoate. It holds a pH below 7.
- **Basic buffer**: a weak base and a salt of that base, for example ammonia and ammonium chloride. It holds a pH above 7.

How an acidic buffer works

The buffer contains large reservoirs of both the weak acid, CH_3COOH, and its conjugate base, CH_3COO^-, which comes almost entirely from the salt: $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$.

- **Add a little acid**: the added H^+ combines with the large supply of ethanoate ions, $CH_3COO^- + H^+ \rightarrow CH_3COOH$, so the equilibrium shifts left and almost all the added H^+ is removed.
- **Add a little alkali**: the OH^- reacts with H^+ to make water, and the equilibrium shifts right as more ethanoic acid dissociates to replace it. The OH^- is also removed directly: $CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$.
- Either way the ratio $[CH_3COOH] / [CH_3COO^-]$ changes only slightly, because both reservoirs are large, so $[H^+]$ and the pH barely change.

Calculating the pH of a buffer

Rearranging K_a: $[\mathrm{H^+}] = K_a \times \dfrac{[\mathrm{HA}]}{[\mathrm{A^-}]}$. Take [HA] as the acid added and $[A^-]$ as the salt added; the tiny amounts that dissociate or react with water make no noticeable difference. Because acid and salt share the same final volume, you can use their **moles** directly in the ratio.

Buffer made by mixing acid and salt

1. 50.0 cm^3 of 0.200 mol dm^{-3} ethanoic acid is mixed with 50.0 cm^3 of 0.100 mol dm^{-3} sodium ethanoate. $K_a = 1.7 \times 10^{-5}$ mol dm^{-3}.
2. Moles of acid = 0.0500 × 0.200 = 0.0100 mol. Moles of ethanoate = 0.0500 × 0.100 = 0.00500 mol.
3. $[\mathrm{H^+}] = 1.7 \times 10^{-5} \times \dfrac{0.0100}{0.00500} = 3.4 \times 10^{-5}\ \mathrm{mol\,dm^{-3}}$, so pH = **4.47**.

Buffer made by partly neutralising a weak acid

1. 50.0 cm^3 of 0.400 mol dm^{-3} ethanoic acid is mixed with 30.0 cm^3 of 0.200 mol dm^{-3} NaOH. Find the pH.
2. Moles of acid = 0.0200 mol; moles of NaOH = 0.00600 mol. The NaOH reacts completely: $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$.
3. Left over: 0.0200 − 0.00600 = 0.0140 mol of acid; formed: 0.00600 mol of ethanoate.

4. $[H^+] = 1.7 \times 10^{-5} \times \dfrac{0.0140}{0.00600} = 3.97 \times 10^{-5}\ \text{mol dm}^{-3}$, so pH = **4.40**.

Adding acid to a buffer

1. 1.00 dm^3 of buffer contains 0.100 mol of ethanoic acid and 0.100 mol of sodium ethanoate, so its pH = pK_a = 4.77. Now 0.0100 mol of HCl is added.
2. The H^+ turns ethanoate into ethanoic acid: acid becomes 0.110 mol, ethanoate becomes 0.0900 mol.
3. $[H^+] = 1.7 \times 10^{-5} \times \dfrac{0.110}{0.0900} = 2.08 \times 10^{-5}\ \text{mol dm}^{-3}$, so pH = **4.68**, a fall of only 0.09.

The same acid added to 1.00 dm^3 of pure water would take the pH from 7.00 to 2.00.

Buffers in blood

Blood plasma must stay between pH 7.35 and 7.45. The main buffer is carbonic acid and hydrogencarbonate ions: $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$. Extra acid from metabolism is taken up by HCO_3^-, shifting the equilibrium left; extra alkali removes H^+, and more H_2CO_3 dissociates to replace it. The carbonic acid level is itself controlled by breathing out carbon dioxide ($H_2CO_3 \rightleftharpoons H_2O + CO_2$). Buffers are also used in foods, shampoos and medicines to keep the pH where the product works best.

Enthalpy change of neutralisation

The **standard enthalpy change of neutralisation** is the enthalpy change when an acid and an alkali react under standard conditions to form one mole of water. For any strong acid with any strong alkali it is almost the same, about −57 kJ mol^{-1}, because both are fully ionised and the only reaction is $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$. With a **weak** acid (or weak base) it is slightly less exothermic, for example about −56 kJ mol^{-1} for ethanoic acid, because some energy is used up dissociating the acid as the reaction proceeds.

Quick checks

QUICK CHECK

In $HSO_4^- + H_2O \rightleftharpoons SO_4^{2-} + H_3O^+$, which is the conjugate base of HSO_4^-?

- H_2O
- H_3O^+
- SO_4^{2-}
- H_2SO_4

HSO_4^- donates a proton and becomes SO_4^{2-}; the pair differs by one H^+. H_3O^+ is the conjugate acid of water.

QUICK CHECK

What is the pH of 0.0150 mol dm^{-3} potassium hydroxide at 298 K?

- 1.82
- 12.18
- 11.82
- 13.18

$[H^+] = 1.00 \times 10^{-14} \div 0.0150 = 6.67 \times 10^{-13}$ mol dm^{-3}, so pH = 12.18. 1.82 is the pOH, which is what you get if you take $-\log[OH^-]$ and stop.

QUICK CHECK

Pure water at 60 °C has pH 6.51. Which statement is correct?

- The water is slightly acidic
- K_w is smaller than at 25 °C
- $[H^+]$ is greater than $[OH^-]$
- The water is neutral and K_w is larger than at 25 °C

In pure water $[H^+]$ always equals $[OH^-]$, so it is neutral. Ionisation of water is endothermic, so K_w rises with temperature and the neutral pH falls below 7.

QUICK CHECK

What is the pH of 0.100 mol dm^{-3} of a weak acid with $K_a = 1.0 \times 10^{-5}$ mol dm^{-3}?

- 3.00
- 1.00
- 6.00
- 5.00

$[H^+] = \sqrt{(1.0 \times 10^{-5} \times 0.100)} = 1.0 \times 10^{-3}$ mol dm^{-3}, so pH = 3.00. 1.00 would be right only for a strong acid. 6.00 comes from forgetting the square root.

QUICK CHECK

Which indicator is suitable for titrating ammonia solution with hydrochloric acid?

- phenolphthalein, because the equivalence point is above 7
- methyl orange, because the steep section is roughly pH 3 to 7
- either, because both change colour near pH 7
- neither, because there is no steep section

A strong acid with a weak base has its steep section in the acidic region and an equivalence point near pH 5, which suits methyl orange. Phenolphthalein would change colour long before the equivalence point.

QUICK CHECK

25.0 cm^3 of a weak acid needs 24.0 cm^3 of NaOH for neutralisation. After 12.0 cm^3 of NaOH has been added the pH is 4.20. What is K_a?

- 4.20 mol dm^{-3}
- 1.6×10^{-2} mol dm^{-3}
- 6.3×10^{-5} mol dm^{-3}
- 4.0×10^{-9} mol dm^{-3}

12.0 cm^3 is half the neutralisation volume, so pH = pK_a = 4.20 and $K_a = 10^{-4.20} = 6.3 \times 10^{-5}$ mol dm^{-3}.

QUICK CHECK

Which mixture is a buffer?

- hydrochloric acid and sodium chloride
- sodium hydroxide and sodium chloride
- ethanoic acid and sodium hydroxide in a 1 : 1 mole ratio
- ammonia and ammonium chloride

A buffer needs a weak acid or weak base together with its conjugate partner. Ethanoic acid and NaOH in a 1 : 1 ratio react completely, leaving only sodium ethanoate; with excess ethanoic acid the mixture would be a buffer.

QUICK CHECK

Why is the enthalpy change of neutralisation of ethanoic acid with NaOH slightly less exothermic than that of HCl with NaOH?

- Ethanoic acid forms less water per mole
- Energy is used to dissociate the weak acid during the reaction
- Ethanoate ions react with water exothermically
- Ethanoic acid has two acidic hydrogens

Both reactions form one mole of water per mole of acid, but ethanoic acid is mostly undissociated, and breaking the O–H bond to release its H^+ takes in some energy.

A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Carbonyls, carboxylic acids and chirality

The C=O group is the busiest functional group in organic chemistry. This topic covers aldehydes, ketones, carboxylic acids and the compounds made from them, the mirror-image molecules that some of their reactions produce, and the instrumental methods (NMR, mass spectrometry and chromatography) that chemists use to work out what they have made.

Naming the compounds in this topic

Family	Functional group	Name ending	Example
aldehyde	–CHO (C=O at the end of the chain)	-al	CH_3CH_2CHO, propanal
ketone	C=O within the chain	-one	$CH_3COCH_2CH_3$, butanone; $CH_3COCH_2CH_2CH_3$, pentan-2-one
carboxylic acid	–COOH	-oic acid	CH_3CH_2COOH, propanoic acid; HOOC–COOH, ethanedioic acid
ester	–COO–	alkyl …-oate	$CH_3COOCH_2CH_3$, ethyl ethanoate
acyl chloride	–COCl	-oyl chloride	CH_3COCl, ethanoyl chloride
amide	$–CONH_2$	-amide	CH_3CONH_2, ethanamide; $CH_3CONHCH_3$, N-methylethanamide
nitrile	–C≡N	-nitrile	CH_3CH_2CN, propanenitrile

- **Aldehydes and carboxylic acids** always have their functional carbon at the end of the chain, so it is carbon 1 and needs no number. In a nitrile, the carbon of the C≡N counts as part of the chain: CH_3CH_2CN has three carbons.
- **Esters** are named from two parts: the alkyl group from the alcohol comes first, then the part from the acid (including the C=O carbon) ending in -oate. $CH_3CH_2COOCH_3$ is methyl propanoate.
- **Hydroxynitriles** are named as nitriles, with the C≡N carbon as carbon 1: $CH_3CH(OH)CN$ is 2-hydroxypropanenitrile.

Chirality and optical isomerism

A carbon atom joined to **four different groups** is a **chiral centre** (an asymmetric carbon). The four groups are arranged tetrahedrally, and there are two ways of doing that which are mirror images of each other. No amount of turning makes one fit exactly on top of the other: they are **non-superimposable mirror images**, like your left and right hands. The two forms are **enantiomers**, or optical isomers.

- **Same chemistry:** enantiomers have identical physical properties and react in the same way with most reagents.
- **Different effect on light:** each enantiomer rotates the plane of **plane-polarised light**, by the same angle but in opposite directions. One turns it clockwise, the other anticlockwise.
- **Racemic mixture (racemate):** a 50 : 50 mixture of the two enantiomers. The rotations cancel, so it has no effect on plane-polarised light: it is optically inactive.
- Draw enantiomers in 3D: two bonds in the plane of the paper, one wedge (towards you) and one dashed (away from you), with the second molecule drawn as the mirror image of the first.

How reaction mechanisms decide optical activity

Whether a reaction gives one enantiomer or a racemic mixture depends on the shape of the particle that the attacking species meets.

- **Attack on a planar group gives a racemate.** A trigonal planar C=O group or carbocation can be attacked equally from above and below. If the new carbon is chiral, the two directions of attack give the two enantiomers in equal amounts. This happens in nucleophilic addition of HCN to aldehydes and most ketones, in the S_N1 substitution of a tertiary halogenoalkane, and in electrophilic addition of HBr to an unsymmetrical alkene such as but-1-ene, where the carbocation is planar.
- **S_N2 inverts the chiral centre.** In S_N2 the nucleophile attacks from the side opposite the leaving group in one step, through a transition state with five groups partly bonded to carbon. The other three groups flip over like an umbrella in the wind, so a single enantiomer of the reactant gives a single enantiomer of the product with the opposite arrangement. The product is optically active.
- **S_N1 goes through a planar carbocation**: the leaving group goes first, then the nucleophile can attack either face, so an optically active reactant gives an optically inactive racemic product. Measuring the optical activity of the product is therefore evidence for which mechanism happened.

Aldehydes and ketones

In the C=O group oxygen is much more electronegative than carbon, so the bond is polar, $C^{\delta+}=O^{\delta-}$. The electron-poor carbon is attacked by **nucleophiles**, unlike the C=C in alkenes, which attracts electrophiles. The C=O bond is also stronger than a C=C bond, so carbonyls do not take part in the electrophilic addition reactions of alkenes.

- **Boiling points:** carbonyl molecules cannot hydrogen bond to each other (no H on O), but they have permanent dipole–dipole forces as well as London forces. Their boiling points are higher than alkanes of similar size but lower than alcohols.

- **Solubility:** small carbonyls such as propanone mix with water, because water molecules hydrogen bond to the lone pairs on the carbonyl oxygen. Solubility falls as the hydrocarbon chain gets longer.

Oxidation: telling aldehydes from ketones

An aldehyde has a hydrogen on its carbonyl carbon and is easily oxidised to a carboxylic acid. A ketone has no such hydrogen, and mild oxidising agents leave it alone. Primary alcohols oxidise to aldehydes and then acids, secondary alcohols to ketones, and tertiary alcohols not at all.

Reagent and conditions	With an aldehyde	With a ketone
potassium dichromate(VI) and dilute sulfuric acid, heat under reflux	orange $Cr_2O_7^{2-}$ turns green (Cr^{3+})	stays orange
Fehling's or Benedict's solution (blue Cu^{2+} complex), warm	brick-red precipitate of copper(I) oxide, Cu_2O	stays blue
Tollens' reagent ($[Ag(NH_3)_2]^+$, made from silver nitrate and aqueous ammonia), warm in a water bath	silver mirror forms on the inside of the tube	no change

Full equations, with ethanal being oxidised to ethanoic acid in each case:

- Dichromate: $3CH_3CHO + Cr_2O_7^{2-} + 8H^+ \rightarrow 3CH_3COOH + 2Cr^{3+} + 4H_2O$ (or simply RCHO + [O] → RCOOH).
- Fehling's: $CH_3CHO + 2Cu^{2+} + 2H_2O \rightarrow CH_3COOH + Cu_2O + 4H^+$
- Tollens': $CH_3CHO + 2Ag^+ + H_2O \rightarrow CH_3COOH + 2Ag + 2H^+$

Reduction with lithium tetrahydridoaluminate

$LiAlH_4$ in dry ether, at room temperature, reduces aldehydes to **primary** alcohols and ketones to **secondary** alcohols. Write the reducing agent as [H]:

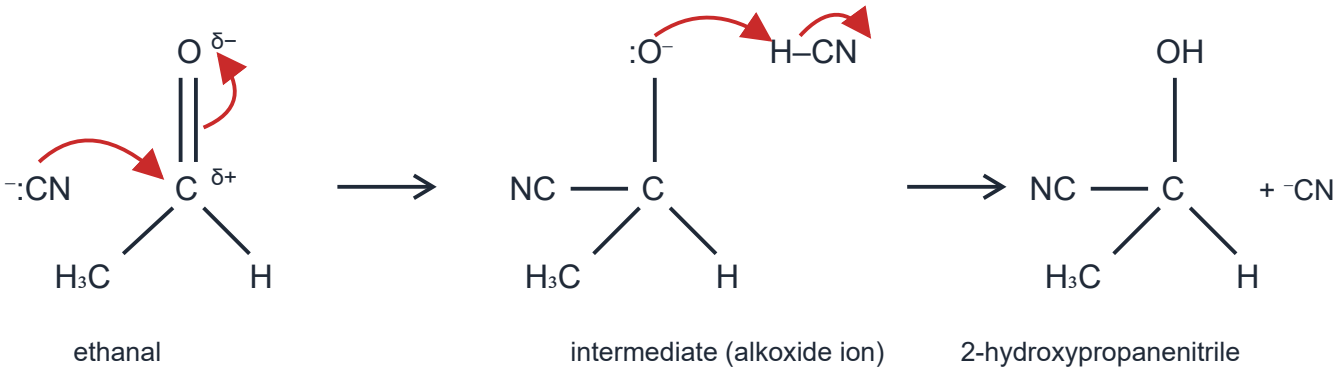
- $CH_3CH_2CHO + 2[H] \rightarrow CH_3CH_2CH_2OH$ (propanal → propan-1-ol)
- $CH_3COCH_3 + 2[H] \rightarrow CH_3CH(OH)CH_3$ (propanone → propan-2-ol)
- The ether must be dry because $LiAlH_4$ reacts violently with water.

Nucleophilic addition of hydrogen cyanide

HCN adds across the C=O bond to give a **hydroxynitrile**, adding one carbon to the chain. The reagent is HCN with some potassium cyanide, at room temperature; HCN itself barely dissociates, so the KCN supplies enough cyanide ions, the actual nucleophile, to start the reaction.

- $CH_3CHO + HCN \rightarrow CH_3CH(OH)CN$ (2-hydroxypropanenitrile)

- CH_3COCH_3 + HCN → $(CH_3)_2C(OH)CN$ (2-hydroxy-2-methylpropanenitrile)



ethanal intermediate (alkoxide ion) 2-hydroxypropanenitrile

Nucleophilic addition. The lone pair on the cyanide ion attacks the δ+ carbon while the C=O π electrons move onto the oxygen. The negative oxygen of the intermediate then takes a proton from HCN, which regenerates a cyanide ion.

KEY IDEA

The carbonyl group is planar, so CN^- can attack from either face. With ethanal, or any unsymmetrical ketone, the product carbon is chiral and the two faces give the two enantiomers equally: the product is a **racemic mixture** with no optical activity.

Two more tests

- **2,4-dinitrophenylhydrazine (2,4-DNPH, Brady's reagent)** gives an orange precipitate with any aldehyde or ketone, but not with other C=O compounds such as acids or esters. Purify the precipitate by recrystallisation and measure its melting point: comparing with tables of known values identifies the exact carbonyl compound.
- **The iodoform test**: warm with iodine and sodium hydroxide. A pale yellow precipitate of triiodomethane (iodoform, CHI_3), with an antiseptic smell, shows a **$CH_3CO–$** group: methyl ketones and ethanal (the only aldehyde that gives it). Alcohols with a $CH_3CH(OH)–$ group also give a positive result, because the alkaline iodine first oxidises them to $CH_3CO–$ compounds.
  $CH_3COCH_3 + 3I_2 + 4NaOH \rightarrow CHI_3 + CH_3COONa + 3NaI + 3H_2O$

Carboxylic acids

- **Hydrogen bonding**: the –COOH group hydrogen bonds strongly, so carboxylic acids have high boiling points. In the pure liquid, and in non-polar solvents, pairs of molecules hydrogen bond into **dimers**, which is why ethanoic acid can appear to have M_r 120 rather than 60.
- **Solubility**: acids with up to four carbons mix completely with water because they hydrogen bond to water molecules; longer chains make them less soluble.
- **Weak acids**: $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$, only slightly dissociated, but still strong enough to release CO_2 from carbonates.

- **Why they are acidic at all**: in the carboxylate ion the negative charge is **delocalised** over both oxygen atoms (the two C–O bonds become the same length). Spreading the charge stabilises the ion, which encourages the O–H bond to break.
- **Effect of substituents**: alkyl groups push electrons towards the COO^- group, concentrating its charge and making the ion less stable, so propanoic acid is slightly weaker than ethanoic acid. Electronegative chlorine atoms pull electron density away, spreading the charge, so chloroethanoic acid is stronger.

Making carboxylic acids

- **Oxidising a primary alcohol or an aldehyde** with excess acidified potassium dichromate(VI), heating under reflux so the aldehyde formed on the way cannot escape: $CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$.
- **Hydrolysing a nitrile** by heating under reflux with dilute hydrochloric or sulfuric acid: $CH_3CH_2CN + 2H_2O + H^+ \rightarrow CH_3CH_2COOH + NH_4^+$. Together with HCN addition, this lets you lengthen a carbon chain and finish with an acid.

Reactions of carboxylic acids

Reagent	Product	Equation
$LiAlH_4$ in dry ether	primary alcohol	$CH_3CH_2COOH + 4[H] \rightarrow CH_3CH_2CH_2OH + H_2O$
reactive metal	salt + hydrogen	$2CH_3COOH + 2Na \rightarrow 2CH_3COONa + H_2$
alkali	salt + water	$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$
carbonate or hydrogencarbonate	salt + water + CO_2	$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$
phosphorus(V) chloride, room temperature	acyl chloride	$CH_3COOH + PCl_5 \rightarrow CH_3COCl + POCl_3 + HCl$
alcohol + concentrated H_2SO_4 catalyst, heat	ester + water	$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

- **Tests for –COOH**: fizzing with sodium carbonate or sodium hydrogencarbonate solution (CO_2 turns limewater milky), and misty fumes of HCl with PCl_5 (PCl_5 also reacts with alcohols, so it shows an –OH group of either kind).
- **Methanoic acid is the exception** that can be oxidised: HCOOH has a structure like an aldehyde (an H on the C=O carbon). It is oxidised to carbonic acid, which breaks down to CO_2 and water, so it gives a positive result with Tollens' and Fehling's.

- **Esterification** is slow and reversible, giving only a modest equilibrium yield; acyl chlorides (below) make esters much better.

Acyl chlorides

Acyl chlorides are far more reactive than carboxylic acids. The carbonyl carbon carries two electronegative atoms, O and Cl, making it strongly δ+, and Cl^- is a good leaving group. They react with nucleophiles at room temperature, quickly and completely, and every reaction gives off steamy fumes of HCl (or makes a chloride salt if a base is present).

Reacts with	Product	Equation
water	carboxylic acid	$CH_3COCl + H_2O \rightarrow CH_3COOH + HCl$
alcohol	ester	$CH_3COCl + CH_3CH_2OH \rightarrow CH_3COOCH_2CH_3 + HCl$
concentrated ammonia	primary amide	$CH_3COCl + 2NH_3 \rightarrow CH_3CONH_2 + NH_4Cl$
primary amine	N-substituted amide	$CH_3COCl + 2CH_3NH_2 \rightarrow CH_3CONHCH_3 + CH_3NH_3^+Cl^-$

With ammonia and amines, the HCl formed is neutralised by a second molecule of the base, so two are needed; with ammonia you see white smoke of ammonium chloride. The reaction with an alcohol is a better way of making esters than using the carboxylic acid, because it is fast and goes to completion.

Esters and polyesters

- **Uses**: many esters smell sweet and fruity, so they are used in perfumes and flavourings (a perfume ester must also be volatile, non-toxic, and unreactive with water). Ethyl ethanoate is a common solvent for glues and inks.
- **Physical properties**: esters are polar but cannot hydrogen bond to each other, so they boil at much lower temperatures than the acids they come from, and they are almost insoluble in water.
- **Acid hydrolysis**: heat under reflux with dilute acid. This is esterification in reverse, so it is reversible and incomplete: $CH_3CH_2COOCH_3 + H_2O \rightleftharpoons CH_3CH_2COOH + CH_3OH$.
- **Alkaline hydrolysis**: heat under reflux with aqueous sodium hydroxide. It goes to completion, giving the salt of the acid and the alcohol: $CH_3CH_2COOCH_3 + NaOH \rightarrow CH_3CH_2COONa + CH_3OH$. The carboxylate ion is not attacked by the alcohol, so the reaction cannot reverse.

Condensation polymers: polyesters

In **condensation polymerisation** monomers join and a small molecule, usually water or HCl, is lost at each link. To build a chain, each monomer needs a reactive group at both ends. A polyester forms from:

- a **dicarboxylic acid and a diol**, losing water at each ester link (slow, needs a catalyst);
- a **diacyl dichloride and a diol**, losing HCl at each link (fast, complete, but the HCl fumes are hazardous);
- a single **hydroxy acid**, whose –OH and –COOH react with neighbouring molecules: 2-hydroxypropanoic acid (lactic acid) gives poly(lactic acid), a biodegradable plastic.

Terylene (PET), used for clothing fibres and drinks bottles, is made from benzene-1,4-dicarboxylic acid and ethane-1,2-diol. Its repeat unit is –OC–C_6H_4–COO–CH_2CH_2–O–. To find the repeat unit of any polyester, take one of each monomer, remove the atoms lost as water or HCl, and join them through the ester link, with bonds left open at both ends.

EXAM TIP

Polyesters contain polar C=O and C–O bonds that can be attacked, so unlike poly(alkenes) they can be **hydrolysed**: by acid, giving back the dicarboxylic acid and the diol, or by alkali, giving the diol and the salt of the acid. This makes them biodegradable.

Spectroscopy: what radiation does to molecules

Radiation	Effect on molecules	Used for
radio waves (in a strong magnetic field)	change the spin state of certain nuclei, such as ^{1}H and ^{13}C	NMR spectroscopy and MRI body scanners
microwaves	make polar molecules rotate faster	heating food, which contains water
infrared	make bonds vibrate more strongly	IR spectroscopy to identify bonds
ultraviolet	can break bonds, for example C–Cl or Cl–Cl, into radicals	starting radical reactions

NMR spectroscopy

Nuclei with an odd number of protons or neutrons, such as ^{1}H and ^{13}C, behave like tiny magnets. In a strong magnetic field they can line up with or against the field, and radio waves of exactly the right frequency flip them between the two. The frequency needed depends on the **chemical environment** of each nucleus: electronegative atoms nearby pull electron density away and shift the signal. This shift is measured on the **δ scale** in parts per million (ppm), relative to a reference.

- **TMS, tetramethylsilane, $Si(CH_3)_4$**, is the reference, set at δ = 0. All twelve of its hydrogens are equivalent, so it gives one strong signal, well away from almost every other signal; it is inert, non-toxic, and has a low boiling point so it is easily removed afterwards.
- **Solvents** must not contain ^{1}H, or they would swamp the spectrum: $CDCl_3$ (D is deuterium, ^{2}H) or CCl_4.

^{13}C NMR

Each different carbon environment gives one peak, so the number of peaks is the number of carbon environments. Carbons near electronegative atoms or in C=O groups appear at higher δ. Symmetry matters: propanone has three carbons but only two environments, so two peaks.

Type of carbon	Approximate δ / ppm
C–C (alkyl)	5 to 40
C–O, C–Cl, C–N	40 to 90
C=C, aromatic carbons	90 to 150
C=O in acids and esters	160 to 185
C=O in aldehydes and ketones	190 to 220

High resolution ^{1}H NMR

A proton NMR spectrum gives four pieces of information:

1. **Number of peaks** = the number of different hydrogen environments. Hydrogens in the same environment are **equivalent**; ethanol, CH_3CH_2OH, has three.
2. **Chemical shift**, δ, tells you what is near each hydrogen (table below).
3. **Relative peak area** (the integration trace) is proportional to the number of hydrogens in each environment: 3 : 2 : 1 for ethanol.
4. **Splitting**: each peak is split by the hydrogens on the **neighbouring** carbon atoms. A peak with *n* neighbouring hydrogens splits into **n + 1** lines (the n + 1 rule). Equivalent hydrogens do not split each other, and O–H protons normally appear as a singlet that does not split its neighbours.

Type of hydrogen	Approximate δ / ppm
R–CH_3	0.7 to 1.2
R–CH_2–R	1.2 to 1.4
R–OH (alcohol)	0.5 to 5.0, variable
H–C next to C=O (H–C–C=O)	2.0 to 2.6
H–C next to O (H–C–O, alcohols, ethers, esters)	3.3 to 4.3
H–C=C (alkenes)	4.5 to 6.0
aldehyde, R–CHO	9.0 to 10.0
carboxylic acid, R–COOH	10.0 to 12.0

Neighbouring H atoms (n)	0	1	2	3	4
Lines in the peak (n + 1)	1, singlet	2, doublet	3, triplet	4, quartet	5, quintet
Relative line heights	1	1 : 1	1 : 2 : 1	1 : 3 : 3 : 1	1 : 4 : 6 : 4 : 1

EXAM TIP

The ethyl group, CH_3CH_2–, is the pattern to recognise: a **triplet** of area 3 (the CH_3, next to a CH_2) and a **quartet** of area 2 (the CH_2, next to a CH_3). A singlet means no hydrogens on the neighbouring carbons, for example a CH_3 joined to a C=O or an O.

Mass spectrometry

In the mass spectrometer an electron is knocked off each molecule, forming the **molecular ion**, M^+, which is both a positive ion and a radical:

$$M \rightarrow [M]^{+\bullet} + e^-$$

The molecular ion peak is the one with the highest m/z (ignoring the small M+1 peak from ^{13}C), and it gives the relative molecular mass. Many molecular ions break up, or **fragment**: a covalent bond breaks to give a positive ion, which is detected, and a radical, which is not: $[M]^{+\bullet} \rightarrow X^+ + Y^\bullet$. The more stable the ion, the taller its peak; acylium ions, $R–C{\equiv}O^+$, and secondary or tertiary carbocations are especially common.

m/z	Common fragment ion
15	CH_3^+
29	$C_2H_5^+$ or CHO^+
43	CH_3CO^+ or $C_3H_7^+$
57	$C_2H_5CO^+$ or $C_4H_9^+$

Telling pentan-2-one from pentan-3-one

1. Both are $C_5H_{10}O$, so both give the molecular ion at m/z 86.

2. Pentan-3-one, $CH_3CH_2COCH_2CH_3$, breaks next to the C=O to give $[CH_3CH_2CO]^+$ at m/z 57 (and $C_2H_5^+$ at 29): $[CH_3CH_2COCH_2CH_3]^{+\bullet} \rightarrow CH_3CH_2CO^+ + \bullet CH_2CH_3$.

3. Pentan-2-one, $CH_3COCH_2CH_2CH_3$, gives a strong peak at m/z 43 from CH_3CO^+ and one at 71 from $CH_3CH_2CH_2CO^+$, but no peak at 57.

The same molecular ion but different fragments: fragmentation patterns tell isomers apart.

Chromatography

Chromatography separates the components of a mixture between a **mobile phase**, which moves, and a **stationary phase**, which does not. Each component is shared between the two phases to a different extent: one that dissolves better in the mobile phase, or is held less strongly by the stationary phase, travels faster. A solid stationary phase separates by **adsorption**; a liquid stationary phase separates by **relative solubility**.

- **Thin-layer chromatography (TLC)**: the stationary phase is a thin layer of silica or alumina on a plate; the mobile phase is a solvent. Draw the start line in **pencil** (ink would dissolve and run), spot tiny amounts of each sample, and stand the plate in a covered tank with the solvent below the start line. Remove it before the solvent reaches the top, mark the solvent front, and show colourless spots with UV light (or ninhydrin spray for amino acids). Wear gloves, so skin oils do not contaminate the plate, and work in a fume cupboard.
- $R_f = \dfrac{\text{distance moved by the spot}}{\text{distance moved by the solvent front}}$. Compare with known substances run on the same plate. Similar compounds may have similar R_f values and not separate.
- **Column chromatography** uses the same principles in a vertical column, so larger amounts can be separated and collected.
- **Gas chromatography (GC)**: the mobile phase is an unreactive carrier gas such as helium or nitrogen; the stationary phase is a high-boiling liquid coated on a solid support inside a long coiled column in an oven. Each component has a characteristic **retention time**, and the area under its peak shows how much is present. It suits volatile liquids.
- **HPLC** (high-performance liquid chromatography): the mobile phase is a liquid forced under high pressure through a column packed with very small solid particles, giving fast, efficient separations.
- **GC–MS**: each component leaving a gas chromatograph is fed straight into a mass spectrometer, and its spectrum is matched against a database. It is used in forensic science, airport security, drug testing and environmental monitoring.

Putting the techniques together

Identifying an unknown compound

1. A compound contains 54.5% carbon, 9.1% hydrogen and 36.4% oxygen by mass. Divide by A_r: C 54.5 ÷ 12 = 4.54, H 9.1 ÷ 1 = 9.1, O 36.4 ÷ 16 = 2.28. Divide by the smallest: C : H : O = 2 : 4 : 1, so the empirical formula is C_2H_4O (mass 44).

2. The mass spectrum has its molecular ion at m/z 88, so the molecular formula is $C_4H_8O_2$.

3. The IR spectrum has a strong absorption at 1740 cm^{-1} (C=O) but no broad absorption at 2500 to 3300 cm^{-1}, so there is no O–H: the compound is an ester, not a carboxylic acid.

4. The ^1H NMR spectrum shows three peaks: δ 4.1, quartet, area 2; δ 2.0, singlet, area 3; δ 1.3, triplet, area 3.

5. The quartet and triplet together are an ethyl group, and the CH_2 at δ 4.1 must be joined to oxygen (O–CH_2CH_3). The singlet of area 3 at δ 2.0 is a CH_3 next to C=O with no neighbouring hydrogens (CH_3CO–).

6. The compound is **ethyl ethanoate**, $CH_3COOCH_2CH_3$. Check: its ^{13}C NMR spectrum would show four peaks, including one near 171 ppm for the ester C=O.

Quick checks

QUICK CHECK

Which of these molecules has a chiral centre?

- propan-2-ol, $CH_3CH(OH)CH_3$
- butan-2-ol, $CH_3CH(OH)CH_2CH_3$
- propanone, CH_3COCH_3
- 2-methylpropan-2-ol, $(CH_3)_3COH$

In butan-2-ol, carbon 2 carries H, OH, CH_3 and C_2H_5: four different groups. In propan-2-ol the two CH_3 groups are the same.

QUICK CHECK

A single enantiomer of 2-bromobutane reacts with OH^- by an S_N2 mechanism. What is the product?

- a racemic mixture of butan-2-ol
- an optically inactive alcohol, because the carbocation is planar
- one enantiomer of butan-2-ol, with the arrangement inverted
- but-2-ene

S_N2 attack comes from the side opposite the bromine in one step, inverting the centre. There is no carbocation; a racemate would point to S_N1.

QUICK CHECK

Which reagent gives a positive result with propanal but not with propanone?

- 2,4-DNPH
- Tollens' reagent
- iodine in sodium hydroxide
- $LiAlH_4$

Only the aldehyde is oxidised, reducing Ag^+ to a silver mirror. 2,4-DNPH reacts with both; the iodoform test is positive for propanone (a methyl ketone) but not propanal.

QUICK CHECK

Why is the product of HCN addition to ethanal optically inactive?

- The product has no chiral centre
- HCN is a weak acid
- The reaction goes by S_N1
- CN^- attacks the planar C=O equally from both sides, forming a racemic mixture

2-hydroxypropanenitrile does have a chiral centre, but both enantiomers form in equal amounts, so their rotations cancel.

QUICK CHECK

Which reaction is the best way to make ethyl ethanoate quickly and completely?

- ethanoic acid + ethanol with a little concentrated H_2SO_4
- ethanoyl chloride + ethanol
- ethanoic acid + ethanal
- ethanoyl chloride + water

Acyl chlorides react with alcohols at room temperature, quickly and irreversibly. Acid-catalysed esterification with the acid is slow and reaches equilibrium.

QUICK CHECK

How many peaks are there in the ^{1}H NMR spectrum of methyl propanoate, $CH_3CH_2COOCH_3$, and what is the splitting of the CH_3 joined to oxygen?

- 2 peaks; triplet
- 3 peaks; quartet
- 3 peaks; singlet
- 4 peaks; singlet

There are three environments: CH_3 of the ethyl group (triplet), CH_2 (quartet) and O–CH_3. The O–CH_3 is a singlet because its neighbour across the oxygen is the C=O carbon, which carries no hydrogen.

QUICK CHECK

Which is NOT a reason for using TMS as the reference in NMR?

- it gives a single strong peak
- it is inert and has a low boiling point
- it dissolves the sample without giving any peaks
- its peak lies away from most other peaks

TMS is added in small amounts as a reference and gives a peak at δ 0. Dissolving the sample without adding peaks is the job of a solvent such as $CDCl_3$.

QUICK CHECK

On a TLC plate the solvent front moved 8.0 cm and a spot moved 2.6 cm. What is its R_f value?

- 3.1
- 0.26
- 0.33
- 5.4

$R_f = 2.6 \div 8.0 = 0.325 \approx 0.33$. R_f is always less than 1; 3.1 is the ratio upside down.